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RESEARCH DISSERTATION

# Characterization of viscoelastic flows in a four-roll mill with physics-informed neural networks

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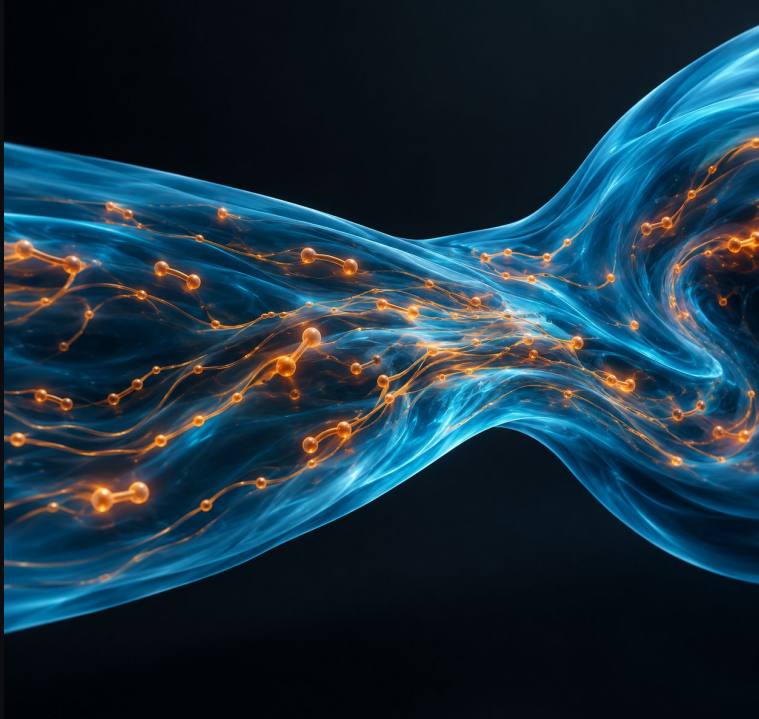
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# The Viscoelasticity Problem: Relevance and Mathematical Complexity



## 01 / FLUID PHYSICS

### Elasticity & Fading Memory

Viscous dissipation, elasticity & deformation history

## 02 / APPLICATIONS

### Technological & Bio Systems

Polymer extrusion, spinning, blood & microfluidics

## 03 / MODELING

### Governing Equations

Non-linear constitutive models (Oldroyd-B, PTT) & mixed nature



# The 4-Roll Mill: Principles & Characteristics

## 01 / KINEMATICS

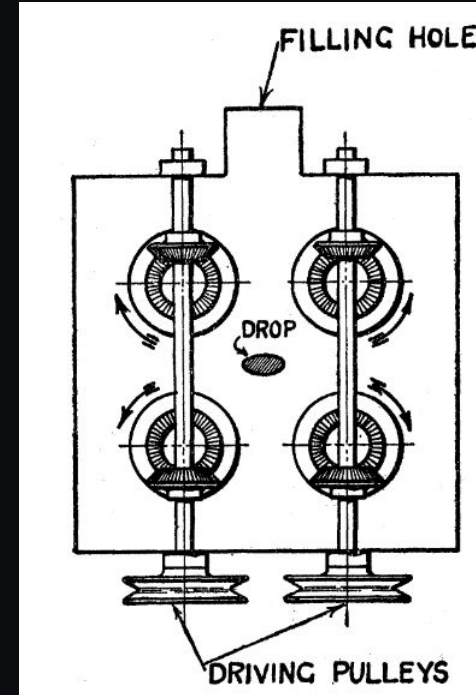
### Independent Roll Rotation

Independent roll speed and direction control to generate varied flow regimes

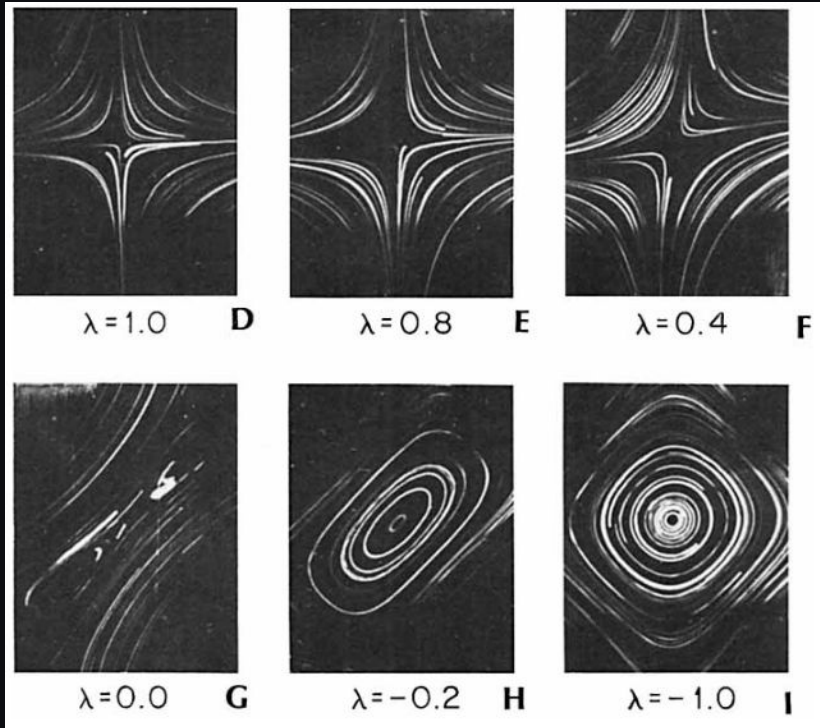
## 02 / CENTRAL REGION

### Stagnation Point

Extensional stagnation point created at the center for fluid characterization.



# 4-Roll Mill Flow Fields



## 01 / FLOW REGIME

### Extensional Flow ( $\lambda=1$ )

Generated when opposing rolls rotate symmetrically, creating pure stretching motion along the central stagnation point.

## 02 / CONTINUOUS SPECTRUM

### Intermediate & Shear Flows

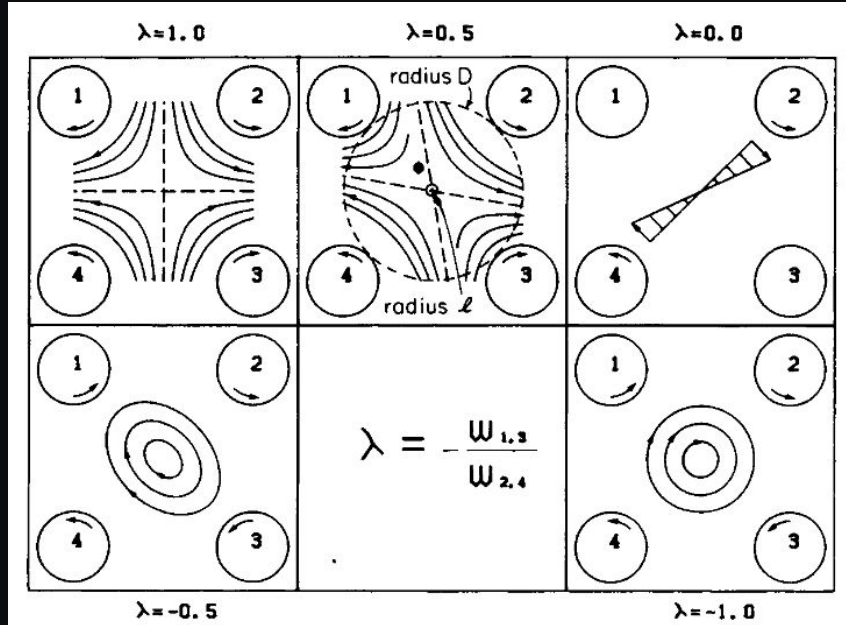
By continuously tuning roll speed ratios, a full spectrum of intermediate regimes is generated, passing through shear flow.

## 03 / FLOW REGIME

### Rotational Flow ( $\lambda=-1$ )

Achieved by driving rolls to induce co-rotational vorticity, placing fluid elements in pure rigid-body rotation.

# 4-Roll Mill Flow Fields



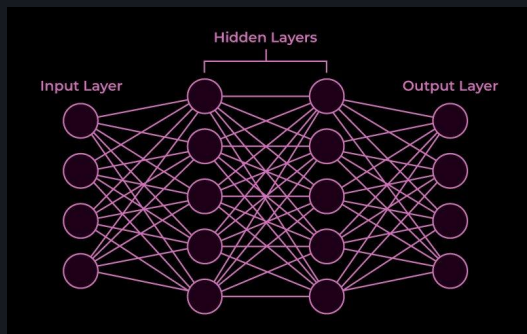
## 01 / KINEMATIC RELATION

$$\omega_1 = \omega_3 = \Omega, \quad \omega_2 = \omega_4 = -\lambda\Omega \quad (-1 \leq \lambda \leq 1)$$

$\Omega > 0$  (strain rate/scale of deformation),  $\lambda$  (flow regime parameter)

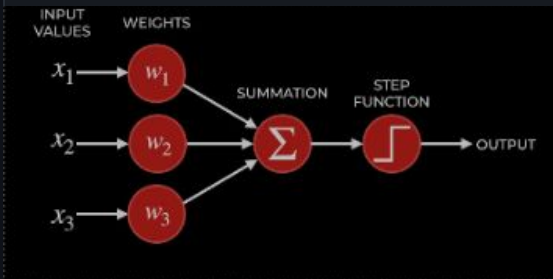
| $\lambda$ | Velocity ( $\omega_1, \omega_2, \omega_3, \omega_4$ ) | Flow Regime        |
|-----------|-------------------------------------------------------|--------------------|
| 1         | $(\Omega, -\Omega, \Omega, -\Omega)$                  | Planar extensional |
| 0         | $(\Omega, 0, \Omega, 0)$                              | Simple shear       |
| -1        | $(\Omega, \Omega, \Omega, \Omega)$                    | Pure rotational    |

# Fundamentals of Deep Learning



01 / ARCHITECTURE

**Fully Connected**



02 / PERCEPTRON

**Activation Function**

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_i^N (y_i - \hat{y}_i)^2$$

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}(\theta)$$

03 / OPTIMIZATION

**Loss & Gradient Descent**

# Why Neural Networks?

## 01 CAPABILITIES

### Cross-Domain Success

Neural networks have demonstrated exceptional problem-solving capabilities across various scientific, engineering, and data-driven domains.

## 02 FUNCTION APPROXIMATION

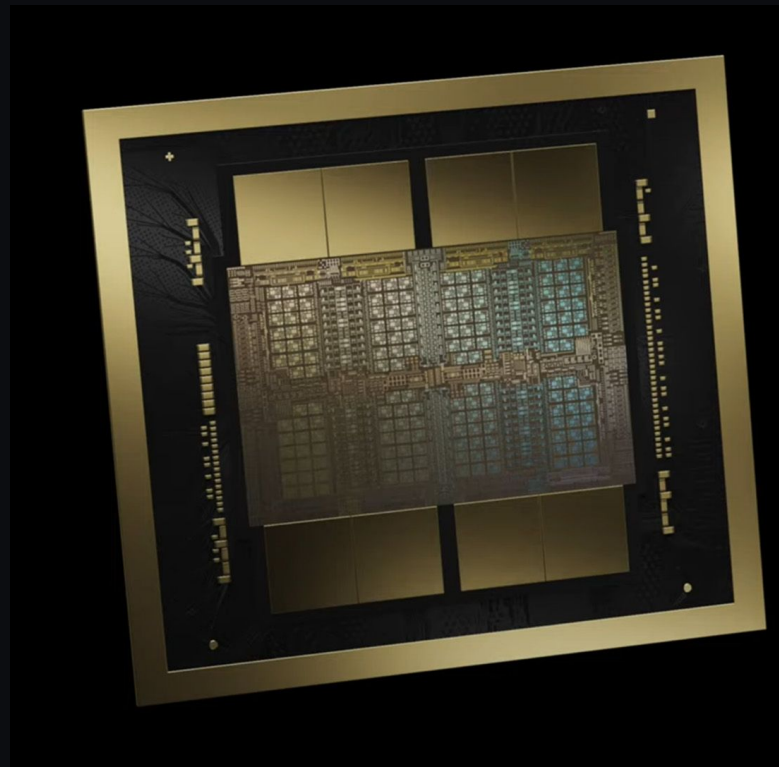
### High-Dimensional Modeling

Acts as universal approximators capable of handling non-linear differential equations and high-dimensional continuous systems without grid constraints.

## 03 HARDWARE SYNERGY

### Computational Progress

Modern computational hardware (GPUs, TPUs) is heavily optimized for parallel tensor operations, aligning directly with deep learning architectures.



NVIDIA Blackwell B200 GPU Die

# Limits of Neural Networks

01

DATA DEPENDENCY

## Limited Data Availability

Standard neural networks rely heavily on large volumes of labeled data, performing poorly or overfitting in data-scarce domains.

02

PHYSICAL CONSISTENCY

## No Physical Grounding

Purely data-driven models lack embedded physical laws, often predicting solutions that violate fundamental physical constraints.

03

TRANSPARENCY

## Black Box Nature

Complex architectures lack interpretability, making it difficult to analyze failure modes or trust outputs in critical applications.

# Physics-Informed Neural Networks (PINNs)

## Hybrid Paradigm

Bridges the structural gap between classical numerical PDE solvers and flexible modern deep learning architectures.

## Loss Regularization

Embeds partial differential equations (PDEs) directly into the neural network's loss function to serve as an exact regularization mechanism.

## Data Efficiency

Drastically reduces the volume of large labeled datasets required for training while enforcing continuous physical consistency.

## Model Explainability

Provides physical interpretability and transparency by grounding neural representations in governing physical laws.



# PINN Framework – Physical Information

$$L_{tot} = L_{data} + L_{physics}$$

## DATA-DRIVEN COMPONENT

### Loss of Data Mismatch

Penalizes difference against sparse experimental observations.

- **Supervised Learning:** Evaluates Mean Squared Error (MSE) at discrete measurement points.
- **Empirical Grounding:** Ensures model predictions conform strictly to observed sensor data & initial/boundary conditions.

## PHYSICS-INFORMED COMPONENT

### Loss of Physics Residuals

Enforces governing PDE laws at unlabelled collocation points.

- **Exact Differentiation:** Computed across domain collocation points via Automatic Differentiation (AD).
- **PDE Regularization:** Penalizes violations of differential conservation laws (e.g. Navier-Stokes).
- **Inverse problem:** Infer the physical parameters that best suit experimental data

# PINN Framework – Automatic Differentiation

## THE ANALOGY

### Backpropagation vs. Coordinate AD

Just as standard gradient descent uses automatic differentiation to compute parameter gradients, it's possible to use computational graph to compute derivatives with respect to spatial inputs.

## EXACT DIFFERENTIATION

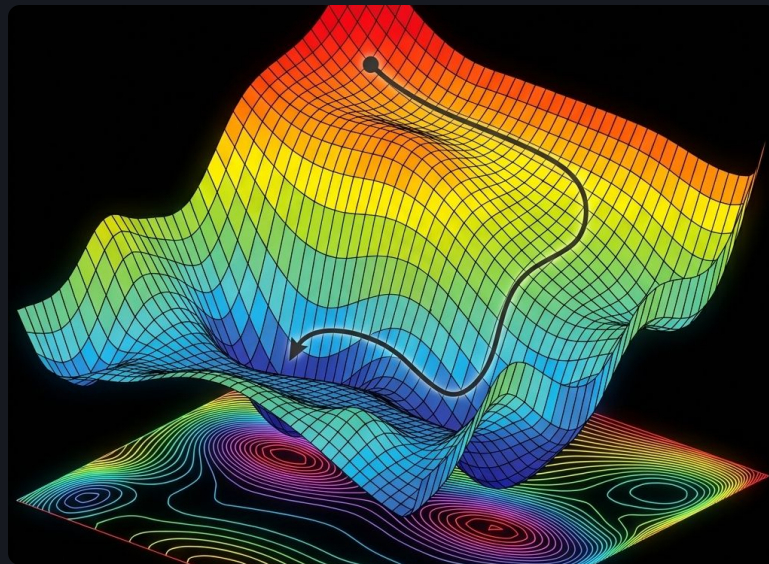
### Pointwise Differential Operators

Evaluates exact PDE residuals ( $\partial u / \partial t, \nabla^2 u$ ) at continuous coordinates without truncation errors, avoiding any finite-difference stencil.

## FUNDAMENTAL PROPERTY

### Meshless Formulation

Completely decouples differential evaluation from geometric topology: collocation points require no cell connectivity, adjacency lists, or domain re-meshing.



## PARAMETER SPACE OPTIMIZATION

Loss surface  $\mathcal{L}(W, b)$  optimized via gradient descent, driving the physics-based PDE residual to zero across all collocation points.

# Navier-Stokes Case Study – Flow Past a Cylinder

Problem Setup, Physical Assumptions & Flow Regime

## FLUID PROPERTIES

### Physical Assumptions

- **Newtonian & Incompressible:** Constant viscosity,  $\nabla \cdot \mathbf{u} = 0$
- **Governing Model:** 2D Navier-Stokes equations

## GEOMETRY & DOMAIN

### 2D Flow Setup

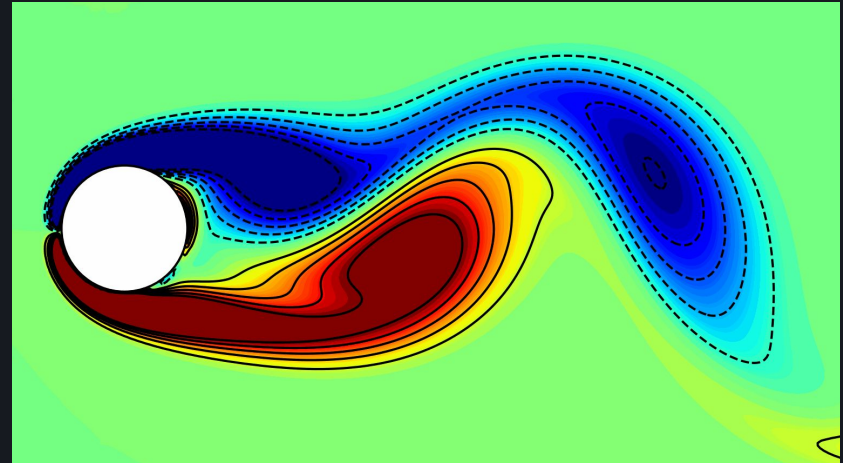
- **Obstacle:** Circular rigid cylinder with downstream wake
- **Sparse Sampling:**  $N = 5,000$  points (1% space-time)

## FLOW REGIME

### Reynolds Number ( $Re = 100$ )

- **Vortex Shedding:** Unsteady Kármán vortex street
- **Inverse Identification:** Parameter recovery ( $\lambda_1, \lambda_2$ )

## FLOW VISUALIZATION



# Navier-Stokes Case Study – Physical vs. PINN Formulation

Inverse Parameter Identification & Physical Model

## REAL PHYSICAL PROBLEM

### Real Navier-Stokes Equations

#### Momentum Balance:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{Re} \left( \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{Re} \left( \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right)$$

#### Continuity Equation:

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

## PINN FORMULATION

### Learned Parameterized Equations

The network learns latent fields  $(u, v, p)$  and concurrently estimates unknown parameters  $\lambda_1$  and  $\lambda_2$ .

#### Residual Loss Formulation:

$$f := u_t + \lambda_1(uu_x + vv_y) + p_x - \lambda_2(u_{xx} + u_{yy}) = 0$$

$$g := v_t + \lambda_1(uv_x + vv_y) + p_y - \lambda_2(v_{xx} + v_{yy}) = 0$$

#### Exact Continuity Satisfaction ( $\psi$ ):

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x} \implies u_x + v_y \equiv 0$$

#### Inverse Parameters Sought by Model:

- $\lambda_1$  (**Convective Term**): Models inertial advective transport.
- $\lambda_2$  (**Diffusive Viscosity**): Corresponds to  $1/Re$  in the Newtonian case.

# Navier-Stokes Case Study – Modeling and Training Parameters

## HYPERPARAMETERS & SETUP

### Model Architecture

- Activation

$\tanh$

- Layers & Neurons

1 network with 9 FC layers  $\times$  20 neurons, 2 outputs

- Optimization

Adam(200k) + L-BFGS (50k)

- Learning Rate

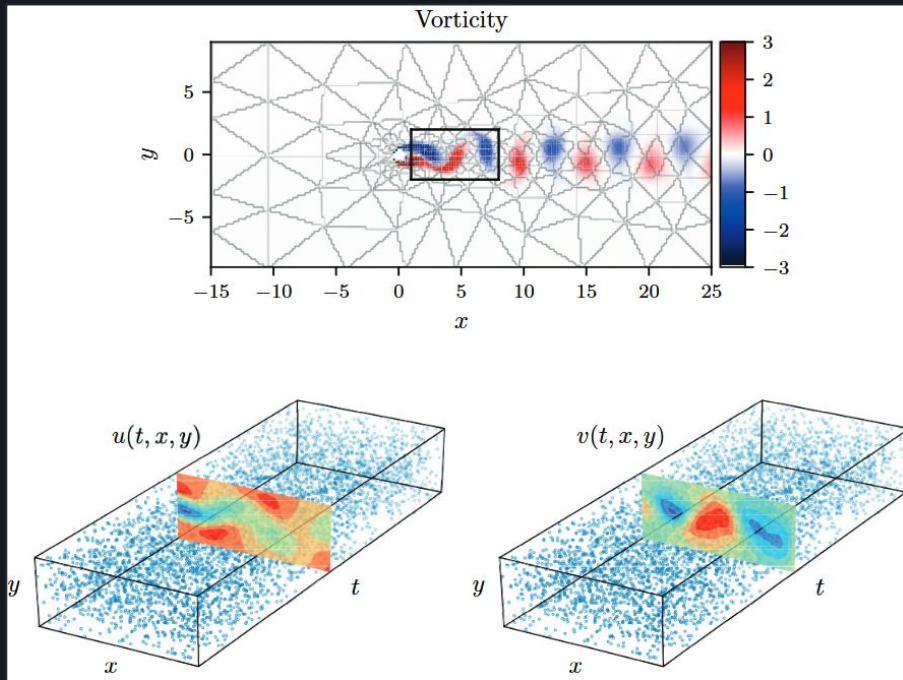
$\eta_0 = 10^{-3}$  fixed

- Sampling Points

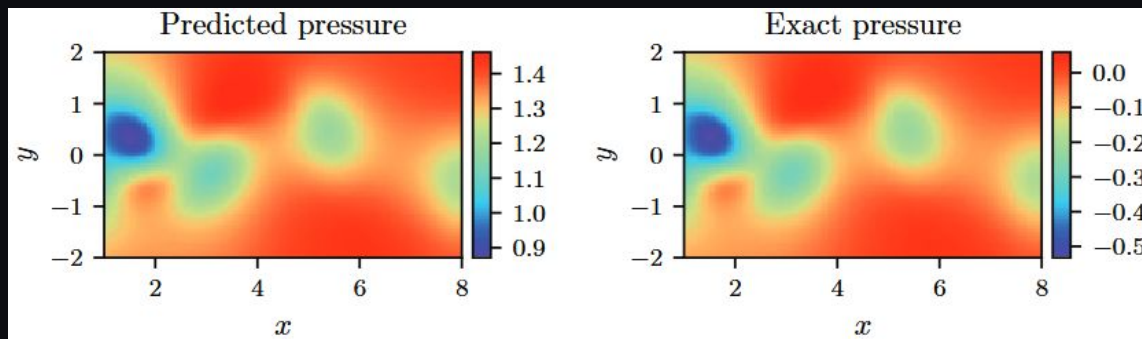
$N = 5,000$  (1% of total data)

## REPRESENTATIVE FLOW FIELDS

### Velocity ( $u, v$ ) data points



# Navier-Stokes Results – Accuracy & Pressure Field



## QUANTITATIVE VALIDATION

### Robust Parameter Identification

Inverse estimation of physical coefficients  $\lambda_1$  and  $\lambda_2$ :

- Clean Data:  $\lambda_1$  error 0.08%,  $\lambda_2$  error 4.67%
- 1% Gaussian Noise:  $\lambda_1$  error 0.17%,  $\lambda_2$  error 5.70%

High stability under experimental measurement noise.

## KEY BREAKTHROUGH

### "Hidden" Pressure Field Reconstruction (p)

- Zero Pressure Training Data: No pressure measurements were supplied during training.
- Physics-Driven Inference: The complete scalar pressure field  $p(t, x, y)$  is inferred strictly via pressure gradient constraints in momentum equations.

*Matches pressure reference up to an additive constant.*

# ViscoelasticNet

## CORE DOMAIN

### Viscoelastic Fluid Modeling

#### KEY CHARACTERISTICS

- Non-Newtonian Flow
- Non-Linear Stresses
- High Weissenberg Instabilities

## INVERSE FORMULATION

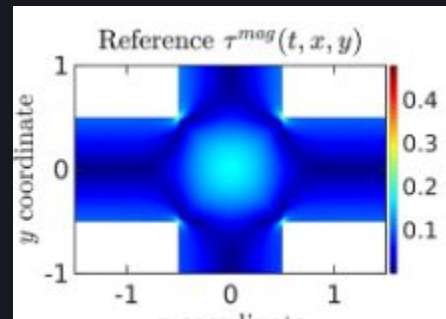
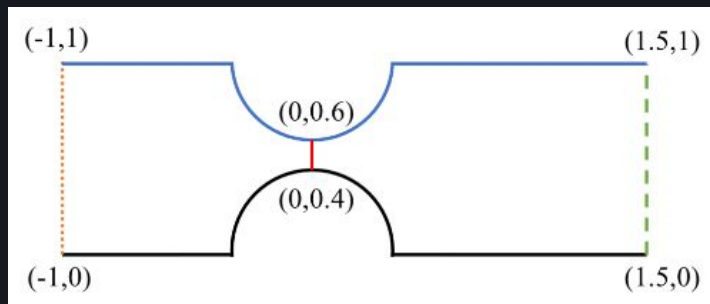
### Parameters & VE Model Selection

#### CAPABILITIES

- **Parameter Estimation:** Learns relaxation time ( $\lambda$ ) and viscosity parameters ( $\eta$ ) directly from data.
- **Constitutive Discovery:** Classifies Oldroyd-B, Giesekus, or PTT model dynamics.
- **Data-Driven:** Reconstructs stress fields from sparse velocity observations.

## BENCHMARK CASE STUDIES

### Stenosis & Cross-Slot Flows



# ViscoelasticNet: Constitutive model(s)

## UNIFIED CONSTITUTIVE FORMULATION

$$\left(1 + \frac{\epsilon \lambda}{\eta_p} \text{tr}(\tau)\right) \tau + \lambda \overset{\nabla}{\tau} + \alpha \frac{\lambda}{\eta_p} (\tau \cdot \tau) = \eta_p (\nabla u + \nabla u^T)$$

$$\overset{\nabla}{\tau} = \frac{D\tau}{Dt} - (\nabla u)^T \cdot \tau - \tau \cdot (\nabla u)$$

$$\frac{D\tau}{Dt} = \frac{\partial \tau}{\partial t} + u \cdot \nabla \tau$$

## MODEL SELECTION CRITERIA

$\epsilon = 0, \alpha = 0$

### Oldroyd-B Model

Standard quasi-linear fluid model with constant viscosity and relaxation time.

$\epsilon = 0, \alpha \neq 0$

### Giesekus Model

Includes non-linear anisotropic drag effects via mobility parameter ( $\alpha$ ).

$\epsilon \neq 0, \alpha = 0$

### Linear PTT Model

Phan-Thien-Tanner model incorporating polymer extensibility parameter ( $\epsilon$ ).

$\epsilon \neq 0, \alpha \neq 0$

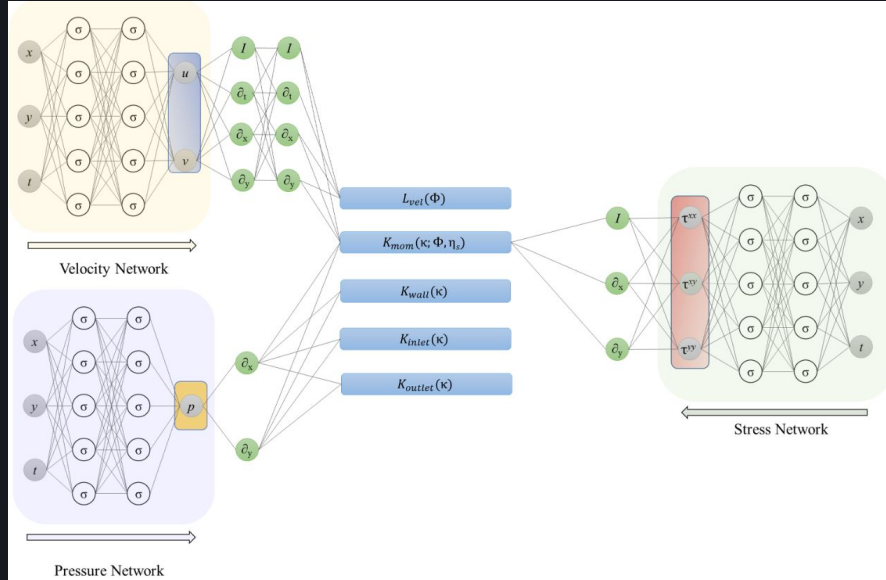
### None of the Three

Model rejected for canonical subset (e.g., validated via *XPomPom* test cases).

# Multi-Network Architecture & Decoupled Optimization

## ARCHITECTURE & PIPELINE

### 3 Dedicated Neural Networks



## DECOUPLED OPTIMIZATION

### Two-Stage Sequential Training

#### STAGE 1: Stress & Rheology

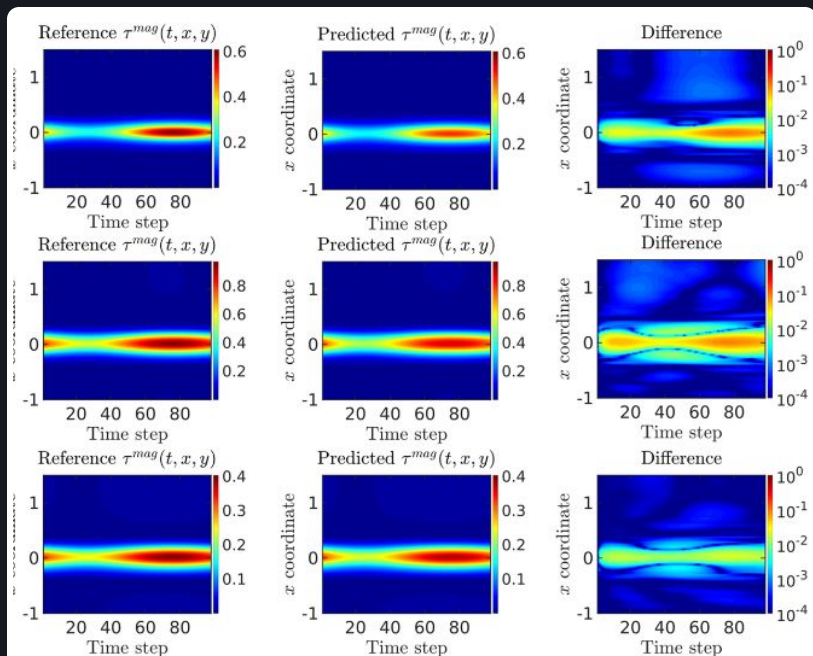
Joint optimization of velocity, stress tensor, and material rheological parameters ( $\lambda, \varepsilon, \alpha, \eta_p$ ) via  $\mathcal{L}_{stress}$  loss.

#### STAGE 2: Pressure & Viscosity

**Freeze** Stress Network weights; train Pressure Network and estimate solvent viscosity  $\eta_o$  by minimizing momentum residual.

# ViscoelasticNet: 2D Stenosis Results

## STRESS FIELD COMPARISON



## PARAMETER IDENTIFICATION UNDER NOISE

|                 | $\alpha$ | $\epsilon$ | $\lambda$ | $\eta_p$ | $\eta_s$ |
|-----------------|----------|------------|-----------|----------|----------|
| Reference value | 0.00     | 0.0        | 0.05      | 0.008    | 0.01     |
| 0% noise        | 0.00     | 0.0        | 0.0517    | 0.0081   | 0.0098   |
| 1% noise        | 0.00     | 0.0        | 0.0517    | 0.0081   | 0.0115   |
| 5% noise        | 0.00     | 0.0        | 0.0517    | 0.0081   | 0.0115   |
| 10% noise       | 0.00     | 0.0        | 0.0517    | 0.0081   | 0.0112   |

|                 | $\alpha$ | $\epsilon$ | $\lambda$ | $\eta_p$ | $\eta_s$ |
|-----------------|----------|------------|-----------|----------|----------|
| Reference value | 0.00     | 0.1        | 0.15      | 0.015    | 0.01     |
| 0% noise        | 0.00     | 0.106      | 0.161     | 0.0157   | 0.0097   |
| 1% noise        | 0.00     | 0.108      | 0.162     | 0.0157   | 0.0123   |
| 5% noise        | 0.00     | 0.108      | 0.162     | 0.0157   | 0.0123   |
| 10% noise       | 0.00     | 0.108      | 0.161     | 0.0157   | 0.0127   |

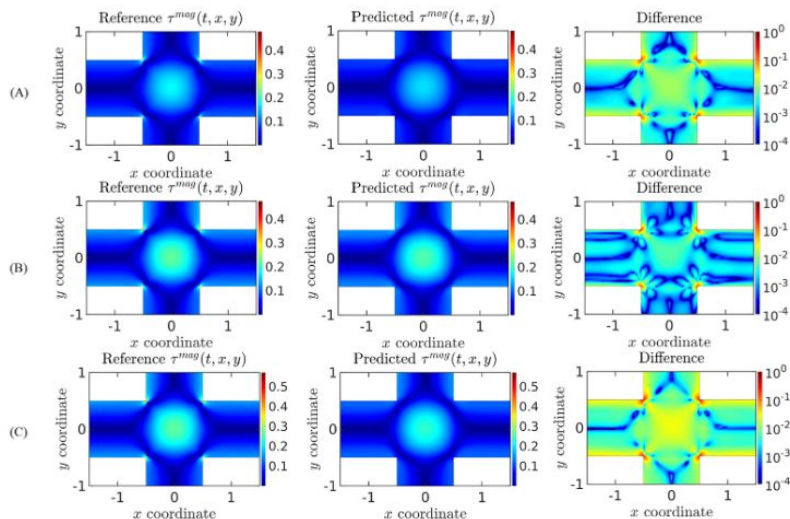
## FIELD ACCURACY & RELATIVE ERROR FORMULATION

|           | $u$                  | $v$                  | $p$                   |
|-----------|----------------------|----------------------|-----------------------|
| 0% Noise  | $1.3 \times 10^{-3}$ | $2.0 \times 10^{-3}$ | $1.95 \times 10^{-1}$ |
| 1% Noise  | $1.4 \times 10^{-3}$ | $2.1 \times 10^{-3}$ | $1.82 \times 10^{-1}$ |
| 5% Noise  | $1.4 \times 10^{-3}$ | $2.2 \times 10^{-3}$ | $1.83 \times 10^{-1}$ |
| 10% Noise | $1.7 \times 10^{-3}$ | $2.5 \times 10^{-3}$ | $1.77 \times 10^{-1}$ |

$$\mathcal{L}(a_{\text{reference}}, a_{\text{prediction}}) = \sqrt{\frac{(a_{\text{reference}} - a_{\text{prediction}})^2}{(a_{\text{reference}} - \bar{a}_{\text{reference}})^2}}$$

# ViscoelasticNet: Cross-Slot Flow Results

## STRESS FIELD COMPARISON



## PARAMETER IDENTIFICATION & PREDICTED VALUES

|                               | $\alpha$ | $\epsilon$ | $\lambda$ | $\eta_p$ | $\eta_s$ |
|-------------------------------|----------|------------|-----------|----------|----------|
| Giesekus Reference value      | 0.05     | 0.0        | 0.004     | 0.003    | 0.01     |
| Giesekus Predicted value      | 0.056    | 0.0        | 0.00386   | 0.0273   | 0.011    |
| Linear PTT Reference value    | 0.0      | 0.02       | 0.008     | 0.025    | 0.01     |
| Linear PTT Predicted value    | 0.0      | 0.0183     | 0.0085    | 0.0245   | 0.0099   |
| Linear PTT Reference value #2 | 0.0      | 0.02       | 0.008     | 0.025    | 0.01     |
| Linear PTT Predicted value #2 | 0.0      | 0.0228     | 0.00836   | 0.0239   | 0.0102   |
| Oldroyd-B Reference value     | 0.0      | 0.0        | 0.005     | 0.01     | 0.01     |
| Oldroyd-B Predicted value     | 0.0      | 0.0        | 0.0046    | 0.0188   | 0.011    |
| Oldroyd-B Reference value #2  | 0.0      | 0.0        | 0.015     | 0.01     | 0.02     |
| Oldroyd-B Predicted value #2  | 0.0      | 0.0        | 0.0135    | 0.0193   | 0.0236   |
| Oldroyd-B Reference value #3  | 0.0      | 0.0        | 0.01      | 0.025    | 0.02     |
| Oldroyd-B Predicted value #3  | 0.0      | 0.0        | 0.0093    | 0.033    | 0.0171   |
| XPomPom Predicted value       | 37.29    | 0.00423    | 0.29      | 0.145    | 0.011    |

# Literature Gap

01

CURRENT STATE

## Simple Benchmarks

Recent developments in PINNs for viscoelastic (VE) fluids have been demonstrated only on simple benchmark problems (such as ViscoelasticNet).

02

THE GAP

## Complex Geometries

There is a need to bridge this gap by extending models to handle complex geometries with realistic, relevant flow features.

03

OBJECTIVE

## Generalization

Developing robust neural network architectures capable of generalizing across diverse physical cases and complex fluid dynamics scenarios.

# From Traditional CFD to Inference: Limits in Inverse Problems

## 01

### THE INVERSE PROBLEM

#### Parameter Estimation

Aims to estimate unknown constitutive parameters (relaxation time  $\lambda$ , solvent/polymer viscosity  $\eta_s, \eta_p$ ).

Inferred directly from experimental field measurements (e.g., PIV velocity fields).

## 02

### FEM / CFD BOTTLENECKS

#### Classical Limitations

- **Iterative Process:** Depends on complex trial-and-error or adjoint sensitivity analysis.
- **Prohibitive Cost:** Requires full nonlinear meshed solves at every optimization step.
- **Numerical Instability:** Divergence at high Weissenberg numbers ( $Wi$ ) breaks optimizers.

## 03

### NEW PARADIGM

#### Physics-Informed NNs

Replaces pure mesh-based solvers with continuous neural representations (PINNs).

Natively incorporates experimental data and differential physical constraints into the loss function for seamless parameter inference.

# Overview of our work

## 01 PARAMETER EXTRACTION

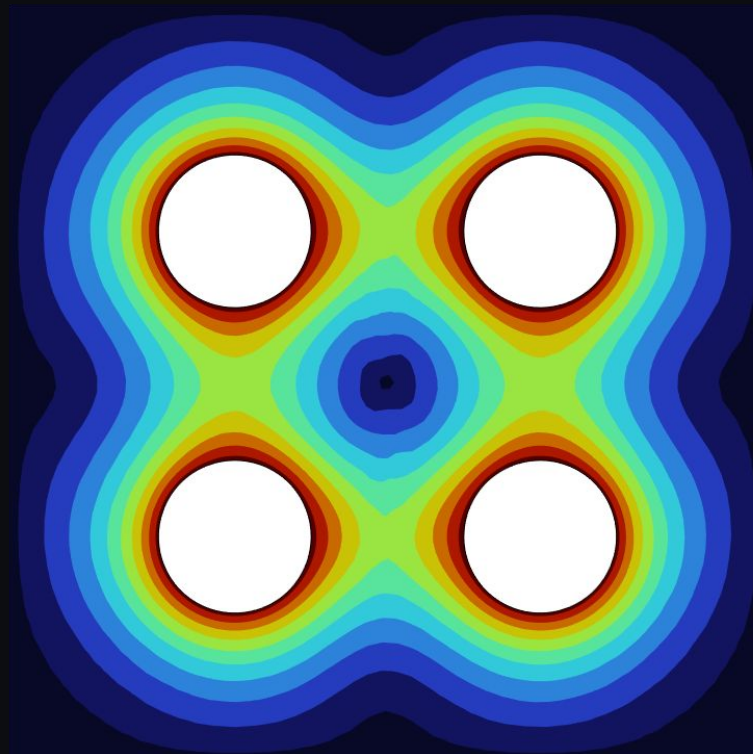
### Physical Parameters

PINN applied to a four-roll mill geometry to extract key physical parameters from experimental fluid dynamics data.

## 02 MODEL SELECTION

### Constitutive Modeling

Identification and evaluation of the constitutive model that best fits and describes experimental observations.



# Differences from ViscoelasticNET

## 01

REGIME TYPE

### Steady-State

Formulated exclusively for stationary conditions  $\rightarrow \partial/\partial t = 0$ .

Neglects transient fluid dynamics to isolate steady flow behavior and optimize computational focus.

## 02

SCOPE & STAGE

### Phase 1 Focus

Restricted to initial baseline configurations  $\rightarrow$  **Phase 1**.

**Constitutive Models:**

Focuses exclusively on **Oldroyd-B** and **Giesekus** models.

## 03

OPERATIONAL BOUNDS

### Extended Ranges

Significantly wider parameter space vs ViscoelasticNET benchmark:

**$\lambda$ : up to 1 s** (vs 0.15 s, **10×**)

**$\eta_p$ : up to 0.98 Pa·s** (vs  $\leq 0.025$  Pa·s, **40×**)

**$\alpha$ : up to 0.50** (vs 0.20)

# ViscoelasticNet: Results

## VALIDATION GEOMETRIES

### Benchmark Flow Cases

#### 1. 2D Stenosis Flow

Hemodynamic pulsatile sinusoidal flow validation:

- **< 5% relative error** on stress fields.
- Accurate classification of Oldroyd-B, Linear PTT, and Giesekus models.

#### 2. Cross-Slot Flow

Strong extensional flow benchmark:

- Correct identification of non-linear rheological parameters.
- Capturing stress singularities at sharp corners requires domain decomposition.

## DATA ROBUSTNESS

### Noise & Sparsity Tolerance

#### Gaussian Noise Resistance

- Tested with up to **10% Gaussian noise** on velocity fields.
- Up to **25% noise** on boundary conditions.
- Learned rheological parameters remain robustly stable.

#### Sparse Data Threshold

- Maintains high accuracy down to **5 × 10<sup>4</sup> total points**.
- Fully compatible with resolutions from real experimental measurements.

## PRACTICAL IMPACT

### Experimental PIV Integration??

#### The PIV Limitation

Laboratory Particle Image Velocimetry (PIV) only provides kinematic velocity fields; internal stresses and pressure are inaccessible directly.

#### Full-Field Virtual Rheometer

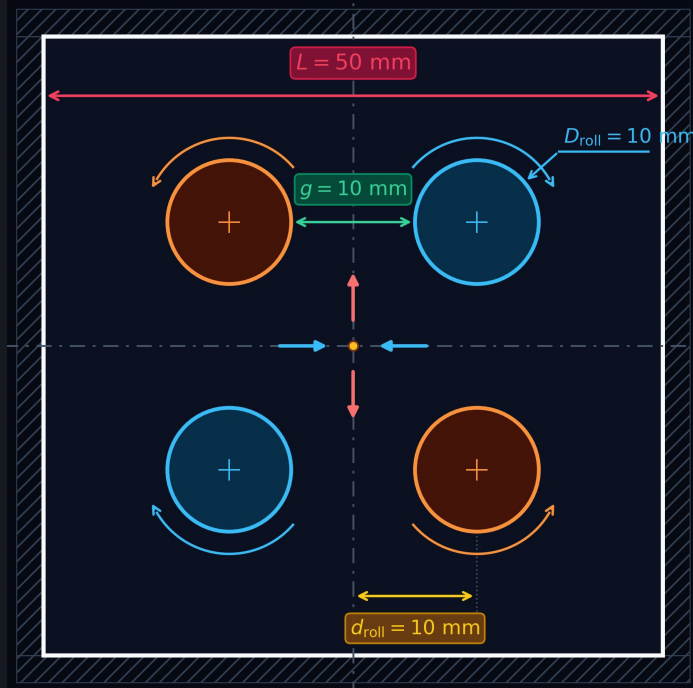
ViscoelasticNet acts as a **virtual rheometer**:

- Uncovers full stress tensors and pressure fields directly from raw PIV data.
- Identifies underlying constitutive equations automatically.

# CFD Modeling: Geometric Ratios & Benchmark Setup

## GEOMETRIC DOMAIN

### Characteristic Dimensions



## GEOMETRIC RATIOS

CYLINDER RATIO ( $D_{\text{roll}} / L$ )

0.2

POSITIONING RATIOS ( $d_{\text{roll}} / L$ )

0.2

# Physical Assumptions

01

FLUID DENSITY

## Incompressible

Constant density  $\rightarrow \nabla \cdot \mathbf{u} = 0$ .

Automatically satisfied through the formulation of the **stream function** ( $\psi$ ).

02

FORCE FIELDS

## No Gravity

Negligible gravity forces  $\rightarrow \mathbf{g} = 0$ .

Isolates purely hydrodynamic forces and eliminates mass/body force terms in momentum balance.

03

THERMODYNAMICS

## Isothermal

Uniform temperature profile  $\rightarrow \mathbf{T} = \text{const}$  ( $\nabla T = 0$ ).

Decouples the energy equation and eliminates thermal dissipation effects.

# Recap: Governing Equations & Boundary Conditions

## 01 CONTINUITY EQUATION

Incompressible Flow Regime

$$\nabla \cdot \mathbf{u} = 0$$

## 02 MOMENTUM BALANCE

Quasi-Stokes Steady-State

$$\rho \left( \frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \eta_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau}$$

## 03 RHEOLOGICAL CONSTITUTIVE EQUATION

Giesekus Model Formulation

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha \lambda}{\eta_p} (\boldsymbol{\tau} \cdot \boldsymbol{\tau}) = 2\eta_p \mathbf{D}$$

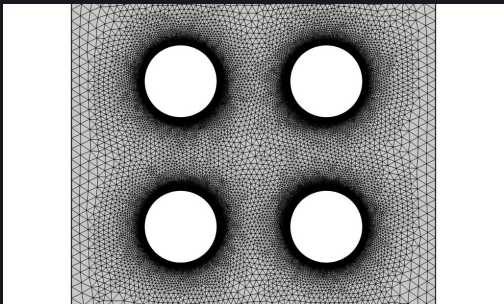
## 04 BOUNDARY CONDITIONS

Wall, Roll No-Slip & Pressure Reference

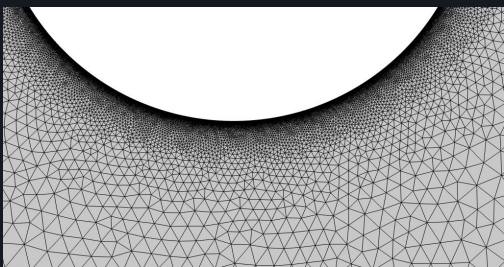
- **Outer Walls (No-Slip):**  $\mathbf{u} = 0$
- **Rotating Rolls(No-Slip):**  $\mathbf{u} = \pm \boldsymbol{\omega} \cdot \mathbf{r}$
- **Pressure Point:**  $p = p_0$

# CFD Meshes & Boundary Detail

125k

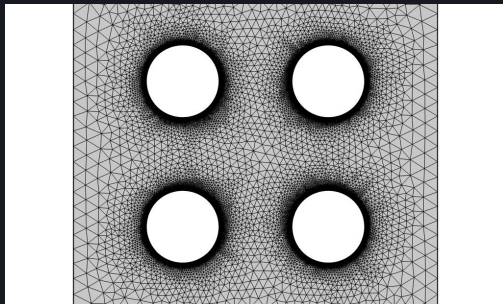


▼ DETAIL

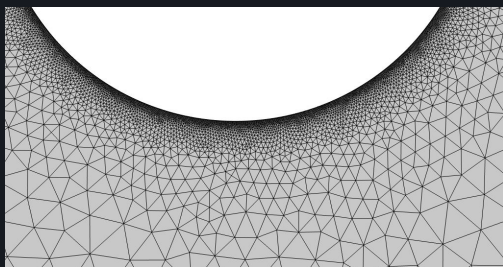


High boundary layer refinement

52k

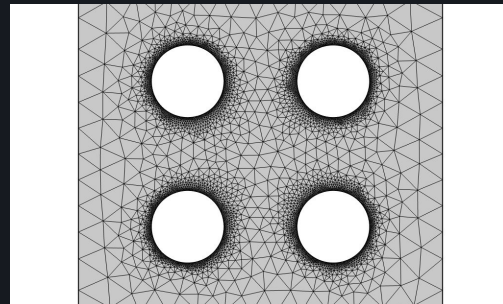


▼ DETAIL

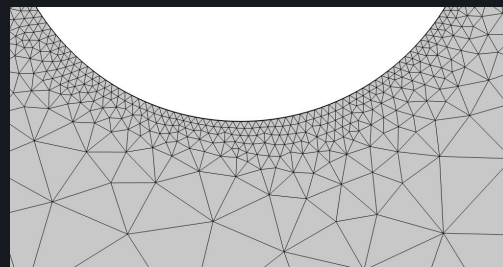


Balanced mesh density

5k



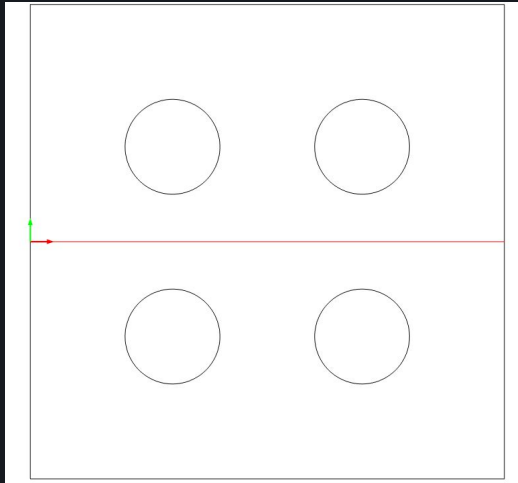
▼ DETAIL



Coarse discretization

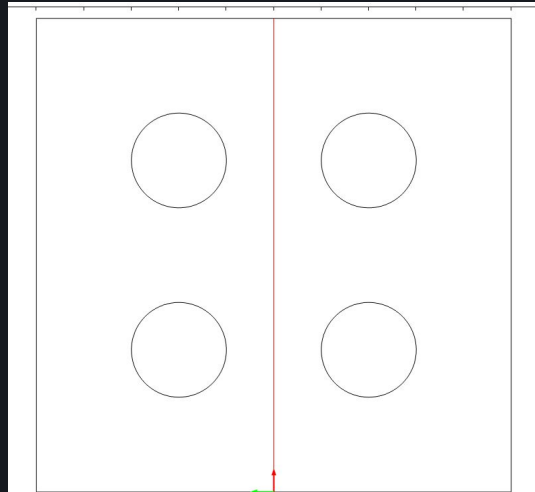
# Relevant Cutlines

**HORIZONTAL CUTLINE**



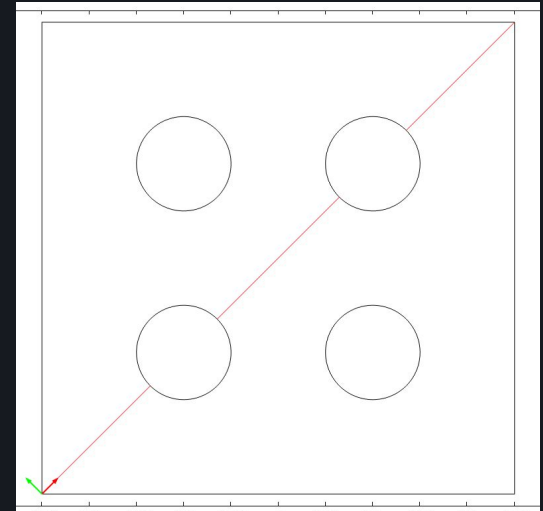
Horizontal cross-section passing through the domain center

**VERTICAL CUTLINE**



Vertical cross-section along the central symmetry axis

**OBLIQUE CUTLINE**



Diagonal cross-section traversing opposite domain corners

# Horizontal Velocity (u)

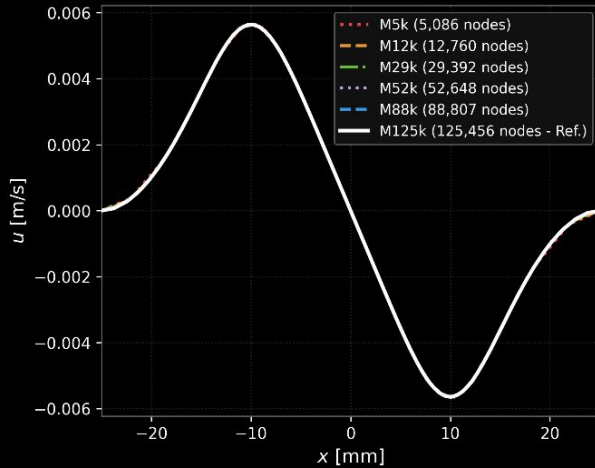
SETUP: Mesh Convergence

MESH: All

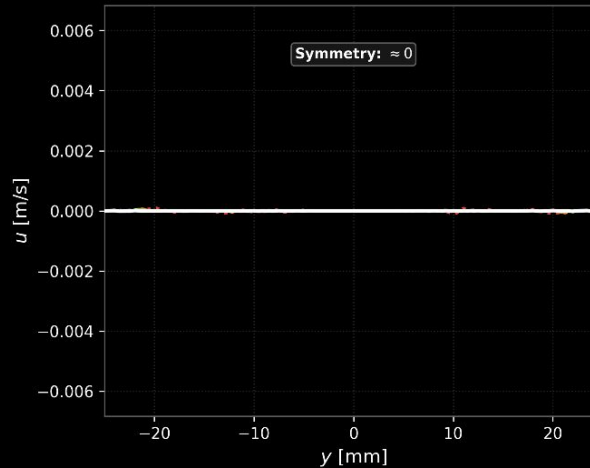
PARAMS:  $\lambda = 1$  s,  $\eta p = 0.9$  Pa s

COMSOL Mesh Convergence Study - Variable:  $u$  [m/s] (Horizontal Velocity)

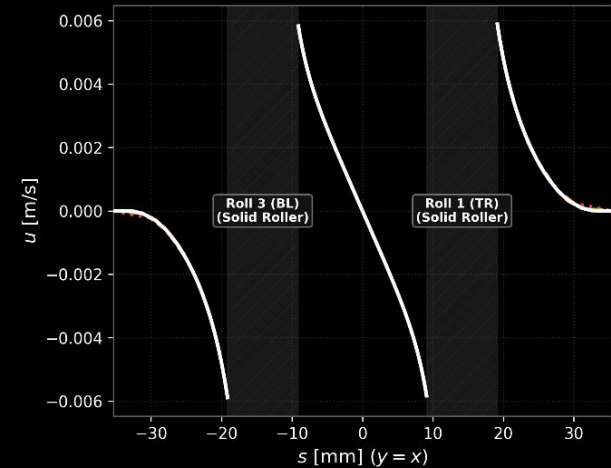
Horizontal Cutline ( $y = 0$ )

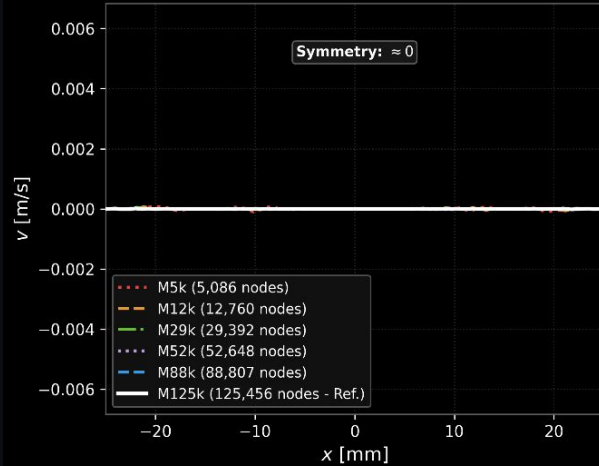
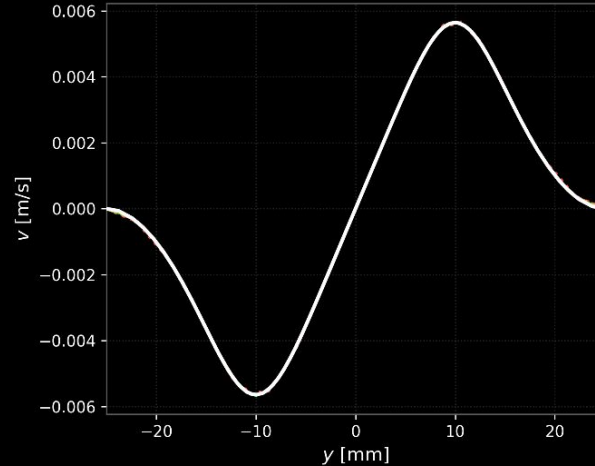
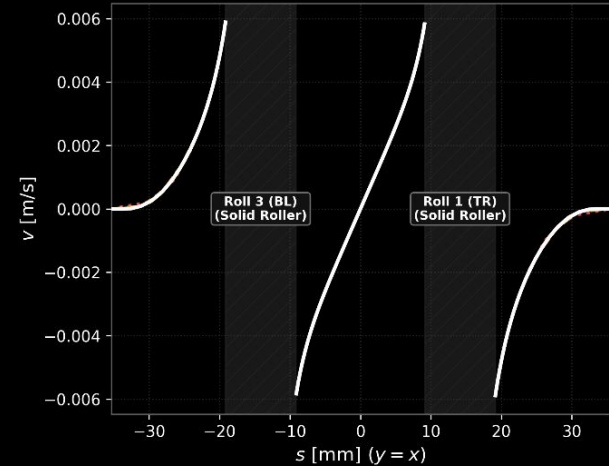


Vertical Cutline ( $x = 0$ )



Diagonal Cutline ( $y = x$ )



**COMSOL Mesh Convergence Study - Variable:  $v$  [m/s] (Vertical Velocity)****Horizontal Cutline ( $y = 0$ )****Vertical Cutline ( $x = 0$ )****Diagonal Cutline ( $y = x$ )**

# Normal extra stress ( $\tau_{xx}$ )

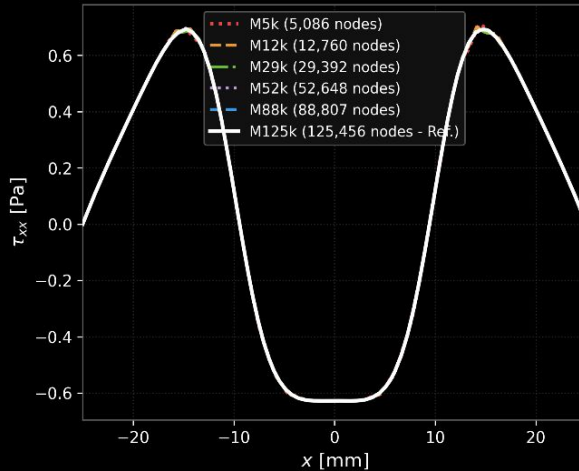
SETUP: Mesh Convergence

MESH: All

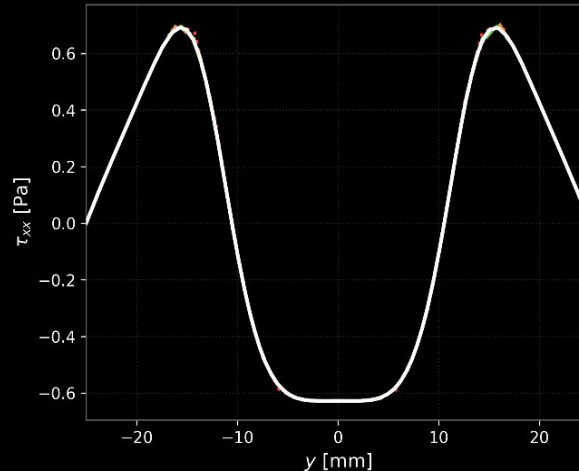
PARAMS:  $\lambda = 1$  s,  $\eta p = 0.9$  Pa s

COMSOL Mesh Convergence Study - Variable:  $\tau_{xx}$  [Pa] (Normal Extra-Stress  $\tau_{xx}$ )

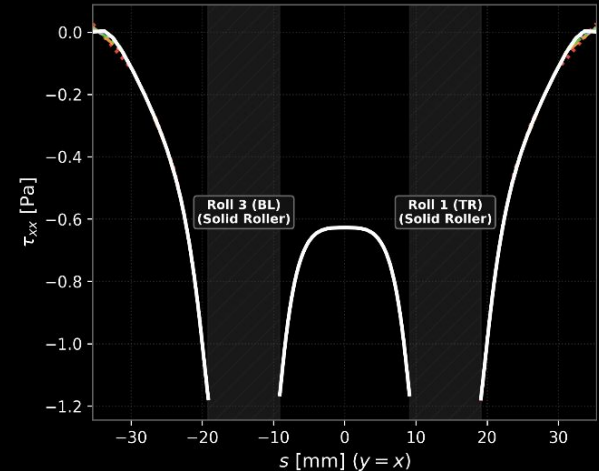
Horizontal Cutline ( $y = 0$ )



Vertical Cutline ( $x = 0$ )



Diagonal Cutline ( $y = x$ )



# Shear stress ( $\tau_{xy}$ )

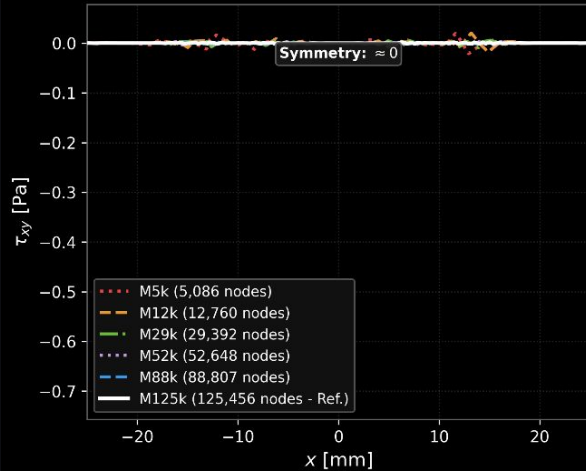
SETUP: Mesh Convergence

MESH: All

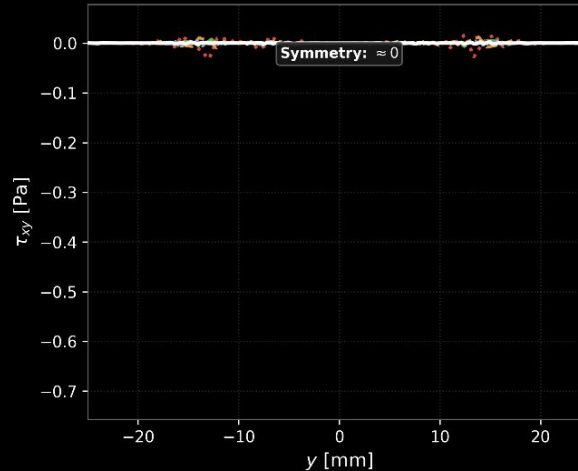
PARAMS:  $\lambda = 1$  s,  $\eta_p = 0.9$  Pa s

COMSOL Mesh Convergence Study - Variable:  $\tau_{xy}$  [Pa] (Shear Extra-Stress  $\tau_{xy}$ )

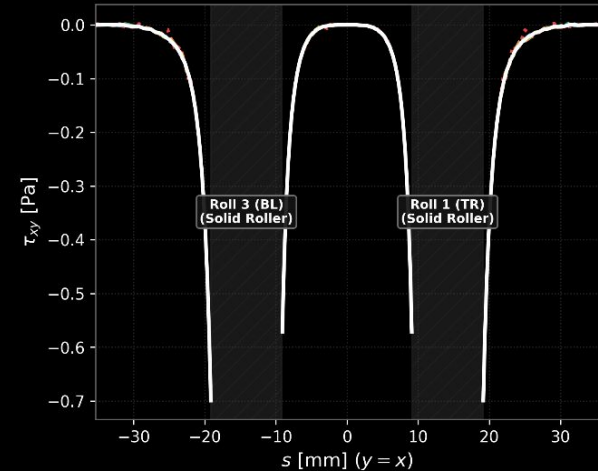
Horizontal Cutline ( $y = 0$ )



Vertical Cutline ( $x = 0$ )

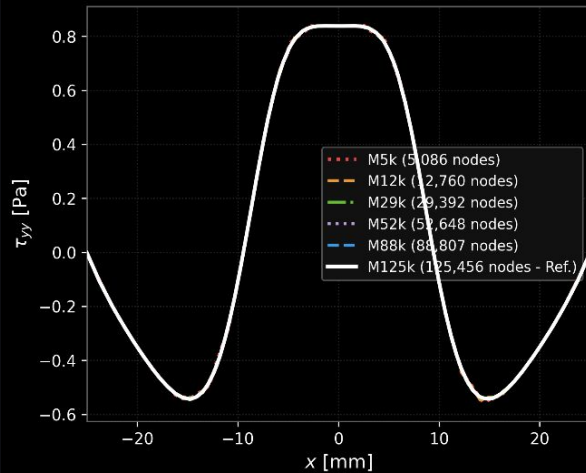


Diagonal Cutline ( $y = x$ )

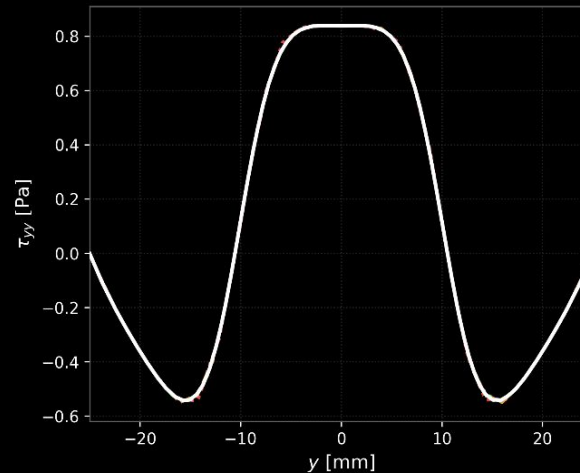


## COMSOL Mesh Convergence Study - Variable: $\tau_{yy}$ [Pa] (Normal Extra-Stress $\tau_{yy}$ )

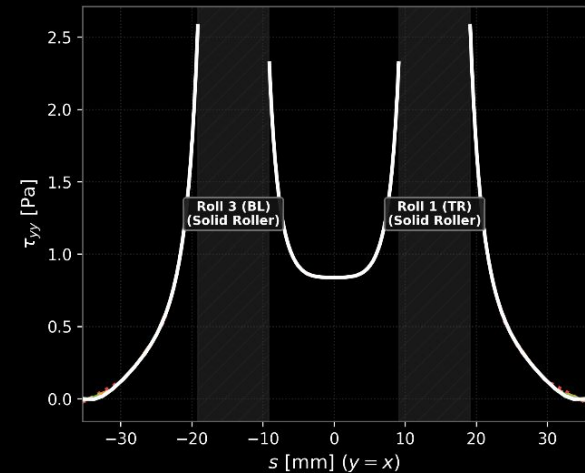
Horizontal Cutline ( $y = 0$ )



Vertical Cutline ( $x = 0$ )



Diagonal Cutline ( $y = x$ )



# Physical parameters & Collocation Points

## Physical Parameters

**Roll Velocity ( $u_{roll}$ ):**  $8.333 \times 10^{-3}$  m/s

**Density ( $\rho$ ):** 1000 kg/m<sup>3</sup>

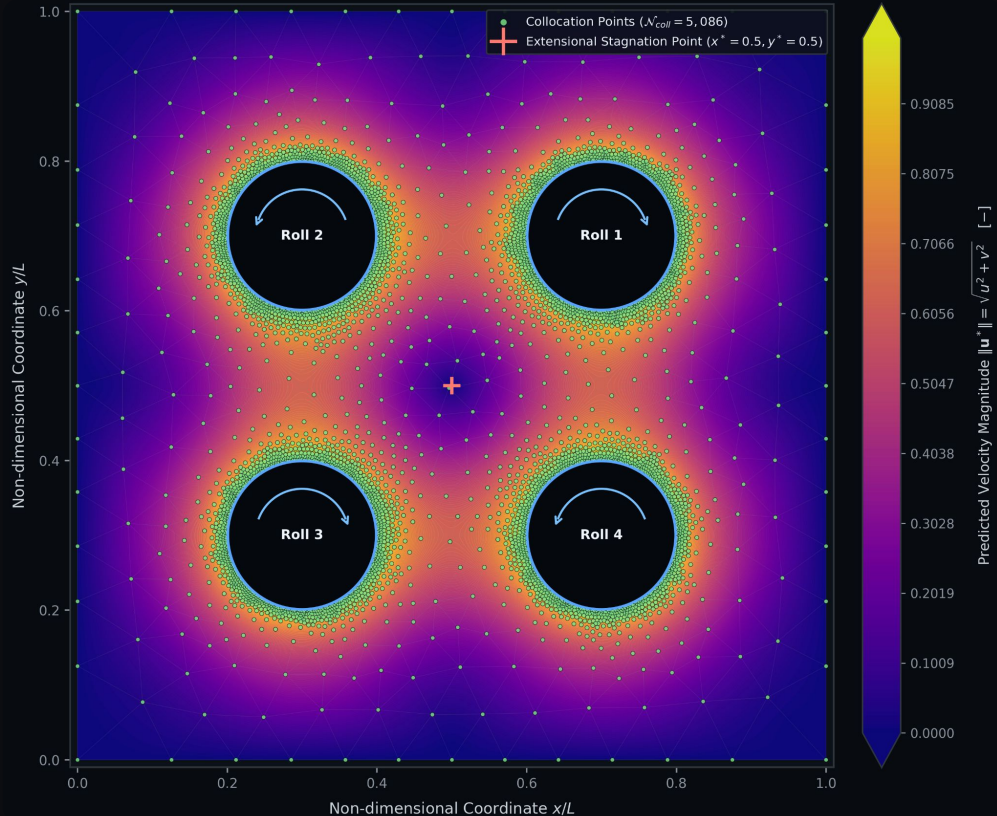
### Flow Characteristics:

Purely extensional flow field considering flow parameter

**Reynolds Num. (Re):**  $0.0417 \ll 1$

**Weissenberg Num. (Wi):**

**Deborah Num. (De):**



# Multi-Objective Loss Function & BC Formulation

$$\mathbf{L_{total}} = w_{data} \cdot \mathbf{L_{data}} + w_{bc} \cdot \mathbf{L_{bc}} + w_{const} \cdot \mathbf{L_{const}} + w_{mom} \cdot \mathbf{L_{mom}}$$

## 1. Velocity Data Loss (Ldata)

$$\mathcal{L}_{data} = \frac{1}{N_u} \sum_{i=1}^{N_u} \|\mathbf{u} - \mathbf{u}_{data}\|^2$$

$$\text{con } \mathbf{u} = \left( \frac{\partial \psi}{\partial y}, -\frac{\partial \psi}{\partial x} \right) \Rightarrow \nabla \cdot \mathbf{u} \equiv 0$$

## 2. Constitutive Eq. (Lconst)

$$\mathbf{r}_{const} = \boldsymbol{\tau} + \lambda \nabla \boldsymbol{\tau} + f_{nl}(\boldsymbol{\tau}) - 2\eta_p \mathbf{D}$$

$$\mathcal{L}_{const} = \frac{1}{N_f} \sum_{i=1}^{N_f} \|\mathbf{r}_{const}\|^2$$

## 3. Momentum Balance (Lmom)

$$\mathbf{r}_{mom} = \rho (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \eta_s \nabla^2 \mathbf{u} - \nabla \cdot \boldsymbol{\tau}$$

$$\mathcal{L}_{mom} = \frac{1}{N_f} \sum_{i=1}^{N_f} \|\mathbf{r}_{mom}\|^2$$

## 4. Boundary Conditions (Lbc)

- No-Slip Kinematics
- Stress Tensor Anchor
- Pressure Point Reference

## 5. Staged Training Strategy

- **Phase 1 (Rheology & Kinematics):**  $w_{mom} = 0$ ,  $w_{const} > 0 \rightarrow$  Trains stream function  $\psi$ , stress  $\tau$ , and learns inverse parameters  $(\lambda, \mu_p, \alpha, \epsilon)$ .
- **Phase 2 (Hydrodynamics & Pressure):**  $w_{const} = 0$ ,  $w_{mom} > 0$ ,  $\tau$  frozen  $\rightarrow$  Solves pressure field  $p$  and solvent viscosity  $\mu_s$  with  $\psi$  fine-tuned at reduced learning rate.

# PINN Training Setup & Parameters [Direct / Forward Problem]

## 1. Network Architecture

Structure: 3 Decoupled Heads

Inputs/Outputs:

$(x, y) \rightarrow \psi(1), p(1), \tau(3: \tau_{xx}, \tau_{xy}, \tau_{yy})$

Layers: 8 hidden layers  $\times$  128 neurons

Activation: **SiLU** (smooth 2nd derivatives)

Weights: Xavier Normal (Zero-init  $p$  &  $\tau$  last layer)

## 2. Staged Optimization Strategy

### PHASE 1: Kinematics & Rheology

- Train  $\psi$  &  $\tau$ ; Freeze  $p$  ( $W_{mom}=0, W_{con}=1.0$ )
- **Fixed Parameters:**  $\lambda, \mu, \alpha, \varepsilon$  (No inverse learning)

### PHASE 2: Hydrodynamics & Pressure

- Freeze  $\tau$  (precomputed  $\nabla \cdot \tau$ ); Train  $p$  &  $\psi$
- **Micro-unfreezing:**  $\psi$  active at LR/10
- **Known solvent viscosity  $\mu_s$**  ( $W_{mom}=1.0, W_{con}=0$ )

## 4. Multi-Stage Precision & Solvers

### Stage 1: Adam

Epochs: 40k (Scratch) / 20k (TL)

Base LR:  $2.5 \times 10^{-3}$

Scheduler: Cosine Annealing  
(down to  $\eta_{min} = 2.5 \times 10^{-6}$ )

Precision: FP32

### Stage 2: L-BFGS

Max Iter: 10k (Scratch) / 5k (TL)

Line Search: **strong\_wolfe**

Precision: Full FP64 / Double  
conversion

## 5. Loss Weights & Infrastructure

### Loss Weights Configuration:

Phase 1:  $W_{data}=1.0$  |  $W_{bc}=5.0$  |  $W_{con}=1.0$  |  $W_{mom}=0.0$  |  $W_{roll\_stress}=1.0$

Phase 2:  $W_{data}=20.0$  |  $W_{bc}=5.0$  |  $W_{mom}=1.0$  |  $W_{con}=0.0$

**Reproducibility:** Fixed Global Seed = 123

# PINN Strategy: Direct vs. Inverse Problem Comparison

## 1. Network Architecture (Shared Baseline)

Structure: 3 Decoupled Heads | Layers: 8 hidden layers  $\times$  128 neurons | Activation: SiLU

Inputs / Outputs:  $(x, y) \rightarrow \psi(1), p(1), \tau(3: \tau_{xx}, \tau_{xy}, \tau_{yy})$  | Epochs: Adam 40k (20kTL)+ L-BFGS 10k (5kTL)

Base LR:  $2.5 \times 10^{-3}$  | Scheduler: Cosine Annealing (down to  $\eta_{\min} = 2.5 \times 10^{-6}$ )

### Direct Problem (Forward)

#### PHASE 1: Kinematics & Rheology

- Train  $\psi$  &  $\tau$ ; Freeze  $p$  ( $W_{\text{mom}}=0, W_{\text{con}}=1.0$ )
- **Fixed Parameters:**  $\lambda, \mu_p, \alpha, \epsilon$  (No inverse learning)

#### PHASE 2: Hydrodynamics & Pressure

- Freeze  $\tau$  (precomputed  $\nabla \cdot \tau$ ); Train  $p$  &  $\psi$
- **Micro-unfreezing:**  $\psi$  active at LR/10
- **Known solvent viscosity  $\mu_s$**  ( $W_{\text{mom}}=1.0, W_{\text{con}}=0$ )

### Inverse Problem (Parameter Learning)

#### PHASE 1: Kinematics & Rheology

- Train  $\psi$  &  $\tau$ ; Freeze  $p$  ( $W_{\text{mom}}=0, W_{\text{con}}=1.0$ )
- **Active Parameters:**  $\lambda, \mu_p, \alpha, \epsilon$  (Inferred online)

**NO PHASE 2**

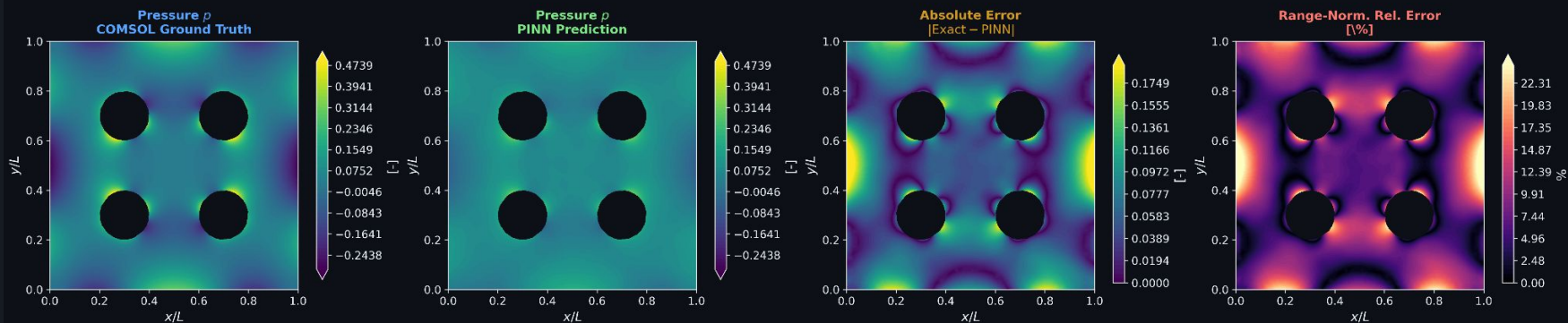
# Four-Roll Mill — Hydrodynamic Pressure ( $p$ )

SETUP: Direct Problem

MESH: M5k

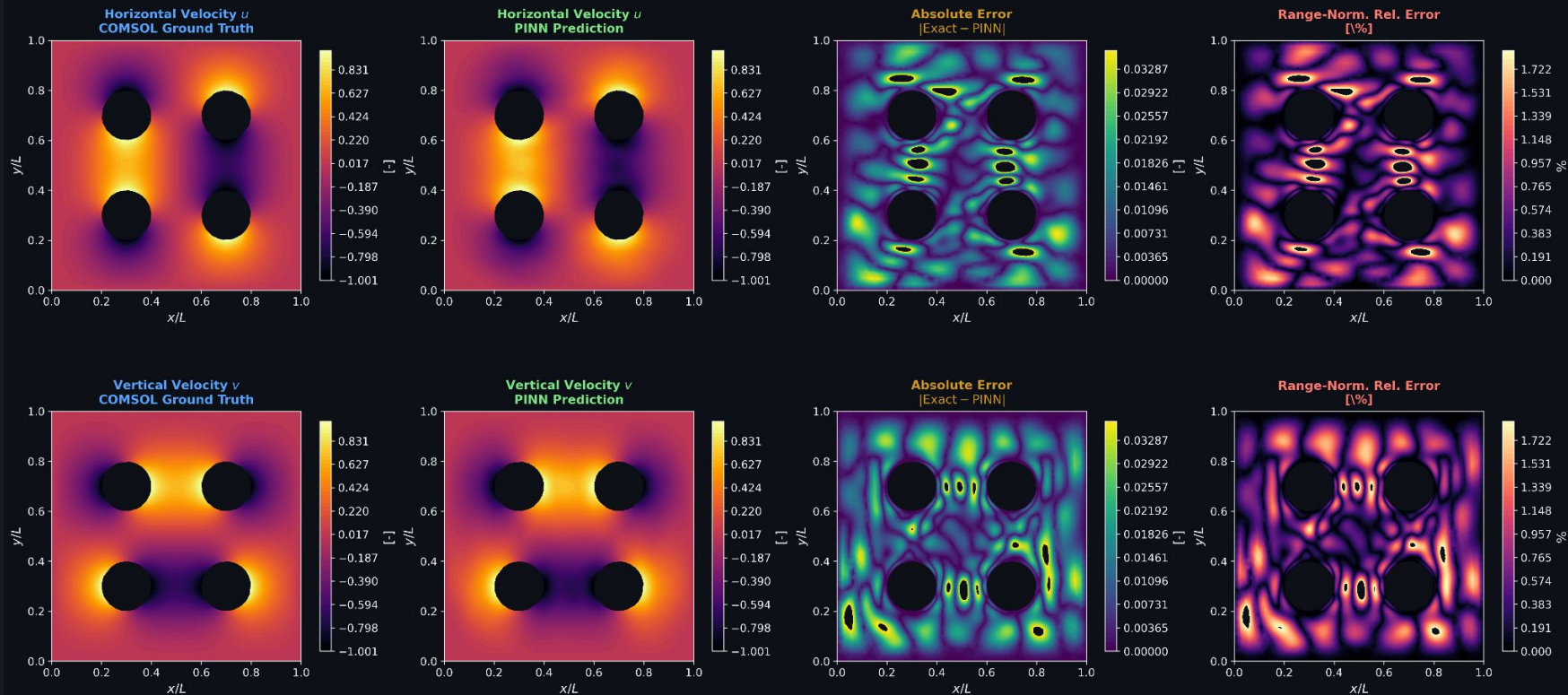
PARAMS:  $\lambda = 0.05$  s,  $\eta p = 0.9$  Pa s

Four-Roll Mill — Direct Benchmark: Hydrodynamic Pressure Field ( $p$ )



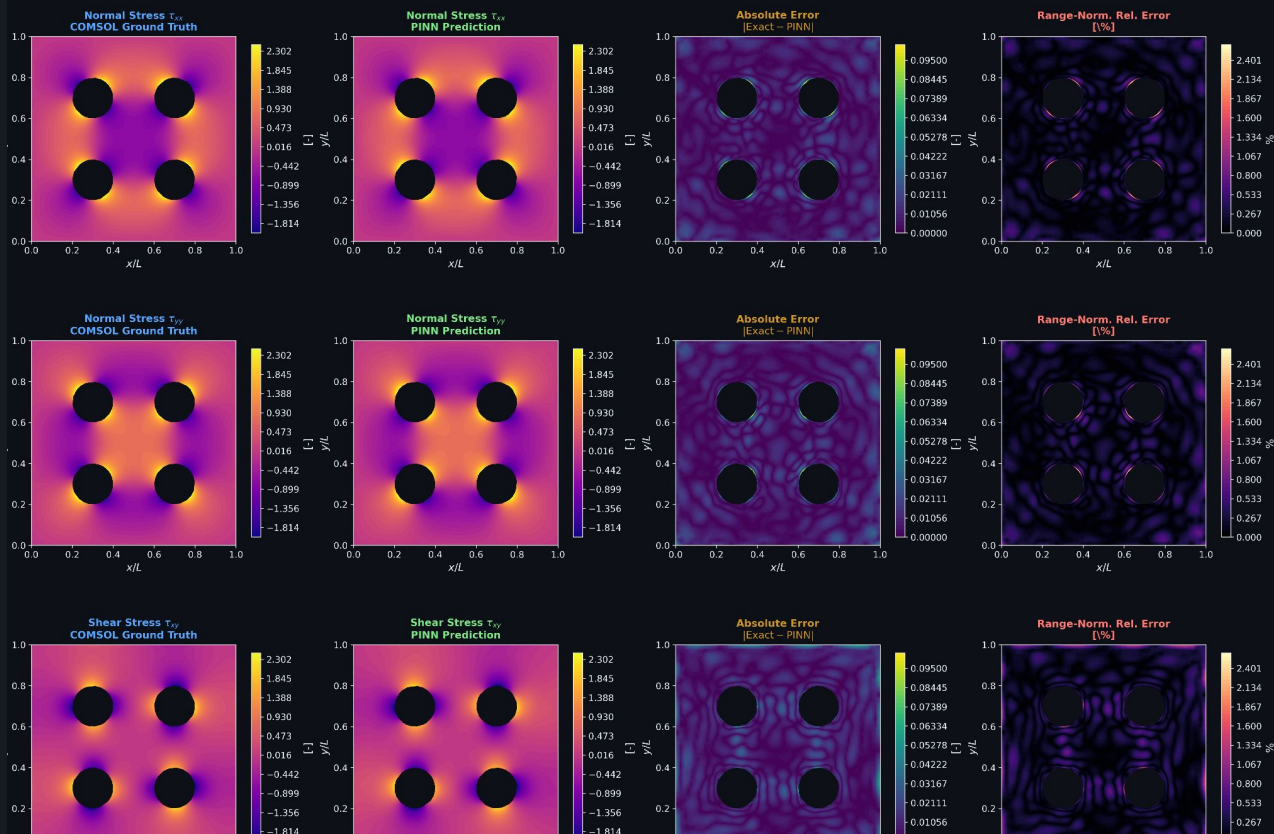
# Four-Roll Mill — Flow Kinematics ( $u, v$ )

Four-Roll Mill — Direct Benchmark: Flow Kinematics ( $u, v$ )



# Four-Roll Mill — Viscoelastic Extra-Stress ( $\tau$ )

Four-Roll Mill — Direct Benchmark: Viscoelastic Extra-Stress ( $\tau$ )



# PINN Training Setup & Parameters [Inverse Problem]

## 1. Network Architecture

Structure: 3 Decoupled Heads

Inputs/Outputs:

$(x, y) \rightarrow \psi(1), p(1), \tau(3: \tau_{xx}, \tau_{xy}, \tau_{yy})$

Layers: 8 hidden layers  $\times$  128 neurons

Activation: **SiLU** (smooth 2nd derivatives)

Weights: Xavier Normal (Zero-init  $p$  &  $\tau$  last layer)

## 2. Staged Optimization Strategy

### PHASE 1: Kinematics & Rheology

- Train  $\psi$  &  $\tau$ ; Freeze  $p$  ( $W_{mom}=0, W_{con}=1.0$ )
- **Active Parameters:**  $\lambda, \mu, \alpha, \varepsilon$

NO PHASE 2

## 4. Multi-Stage Precision & Solvers

### Stage 1: Adam

Epochs: 40k (Scratch) / 20k (TL)

Base LR:  $2.5 \times 10^{-3}$

Scheduler: Cosine Annealing  
(down to  $\eta_{min} = 2.5 \times 10^{-6}$ )

Precision: FP32

### Stage 2: L-BFGS

Max Iter: 10k (Scratch) / 5k (TL)

Line Search: **strong\_wolfe**

Precision: Full FP64 / Double  
conversion

## 5. Loss Weights & Infrastructure

### Loss Weights Configuration:

Phase 1:  $W_{data}=1.0$  |  $W_{bc}=5.0$  |  $W_{con}=1.0$  |  $W_{mom}=0.0$  |  $W_{roll\_stress}=1.0$

Phase 2:  $W_{data}=20.0$  |  $W_{bc}=5.0$  |  $W_{mom}=1.0$  |  $W_{con}=0.0$

**Reproducibility:** Fixed Global Seed = 123

# Four-Roll Mill — Inverse Problem Results

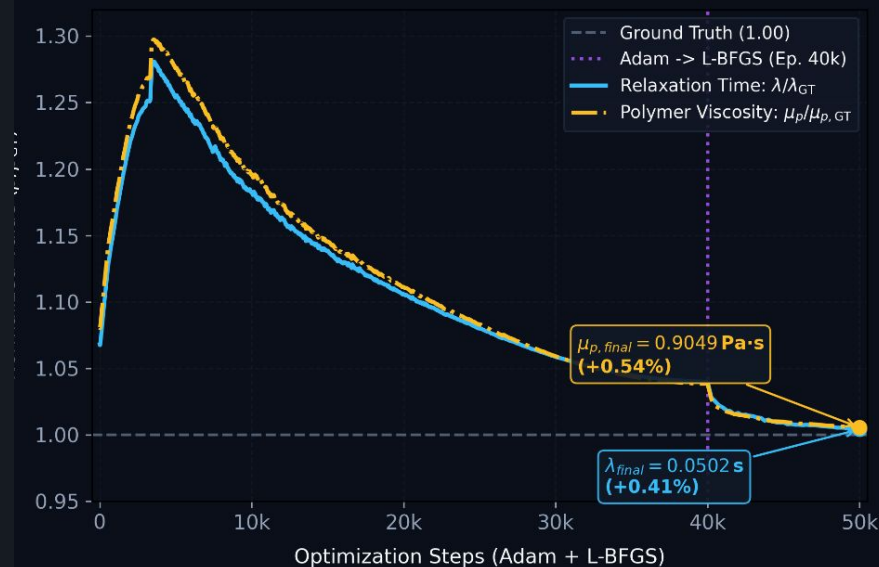
SETUP: Inverse Parameter Estimation

MESH: M125k (Highest Resolution)

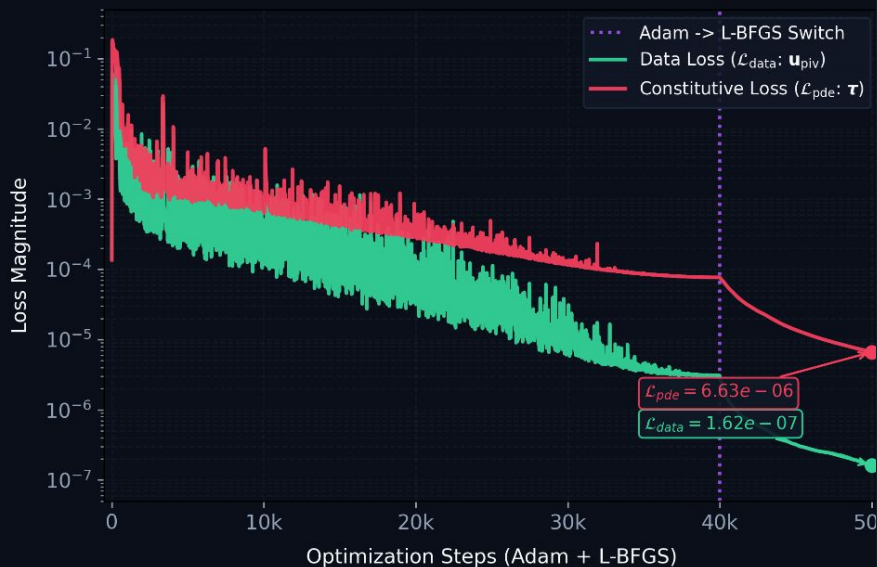
PARAMS:  $\lambda = 0.05$  s,  $\eta_p = 0.9$  Pa s

Four-Roll Mill Inverse Benchmark — M125k Asymptotic Mesh (Oldroyd-B:  $\lambda = 0.050$  s,  $\mu_p = 0.900$  Pa·s)

A. Normalized Parameter Convergence



B. Objective Function Decay (Log-Scale)



# Inverse Problem: Key Parameter Discovery Benchmarks

| Case # | Constitutive Model & Setup                                            | Training Strategy & TF Source                                                                   | Target Parameter                                                                                     | Ground Truth               | PINN Prediction            | Parameter Error            | Velocity Error $L_2$ |
|--------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|----------------------------|----------------------------|----------------------------|----------------------|
| Case 1 | Canonical Oldroyd-B<br>(Standard Low- $Wi = 0.08$ )                   | Ex-Novo (From Scratch)<br>40k Adam + 10k L-BFGS<br>Donor: None                                  | Relaxation time $\lambda$ [s]<br>Polymer viscosity $\mu_p$ [Pa·s]                                    | 0.0500<br>0.9000           | 0.0502<br>0.9049           | +0.41%<br>+0.54%           | 0.040%               |
| Case 2 | Oldroyd-B High Elasticity<br>(High- $Wi = 1.17$ , Boger Fluid)        | Transfer Learning (TF)<br>20k Adam + 5k L-BFGS (-50% budget)<br>Donor: ckpt_L0.2-P0.5-S0.5      | Relaxation time $\lambda$ [s]<br>Polymer viscosity $\mu_p$ [Pa·s]                                    | 0.7000<br>0.5000           | 0.7089<br>0.5019           | +1.27%<br>+0.38%           | 0.330%               |
| Case 3 | Nonlinear Giesekus Model<br>(Blind guess $\alpha_0 = 0.25$ )          | Ex-Novo (From Scratch)<br>40k Adam + 10k L-BFGS<br>Donor: None                                  | Mobility parameter $\alpha$ [-]<br>Relaxation time $\lambda$ [s]<br>Polymer viscosity $\mu_p$ [Pa·s] | 0.3500<br>0.1000<br>0.5000 | 0.3298<br>0.1032<br>0.5307 | -5.76%<br>+3.16%<br>+6.14% | 0.460%               |
| Case 4 | Giesekus Extreme Mobility<br>(Strong shear-thinning $\alpha = 0.50$ ) | Transfer Learning (TF)<br>20k Adam + 5k L-BFGS (-50% budget)<br>Donor: ckpt_giesekus_L0.1-A0.35 | Mobility parameter $\alpha$ [-]<br>Relaxation time $\lambda$ [s]<br>Polymer viscosity $\mu_p$ [Pa·s] | 0.5000<br>0.3000<br>0.0500 | 0.4665<br>0.3077<br>0.0522 | -6.70%<br>+2.57%<br>+4.40% | 0.520%               |

# Grid Convergence Study — Accuracy vs. Mesh Resolution

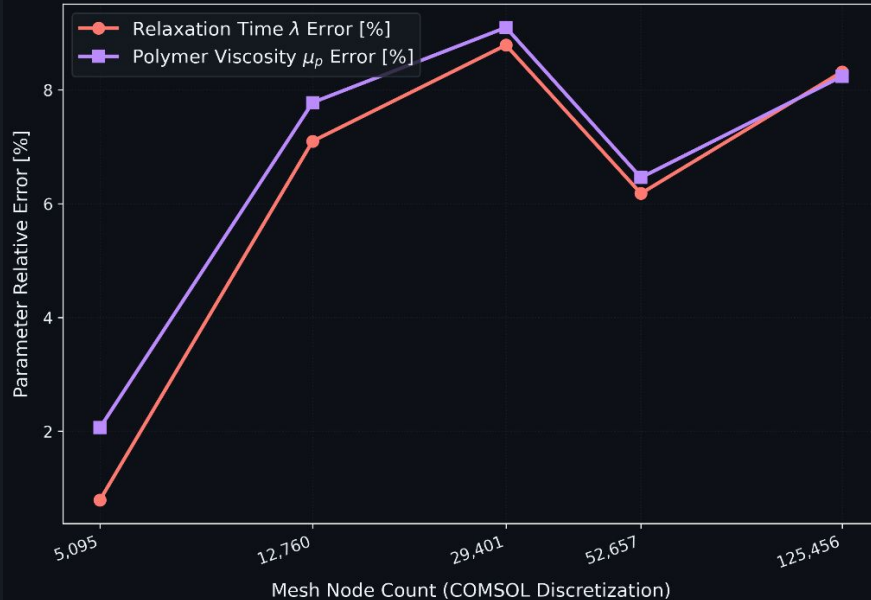
SETUP: Mesh Robustness

MESH: All

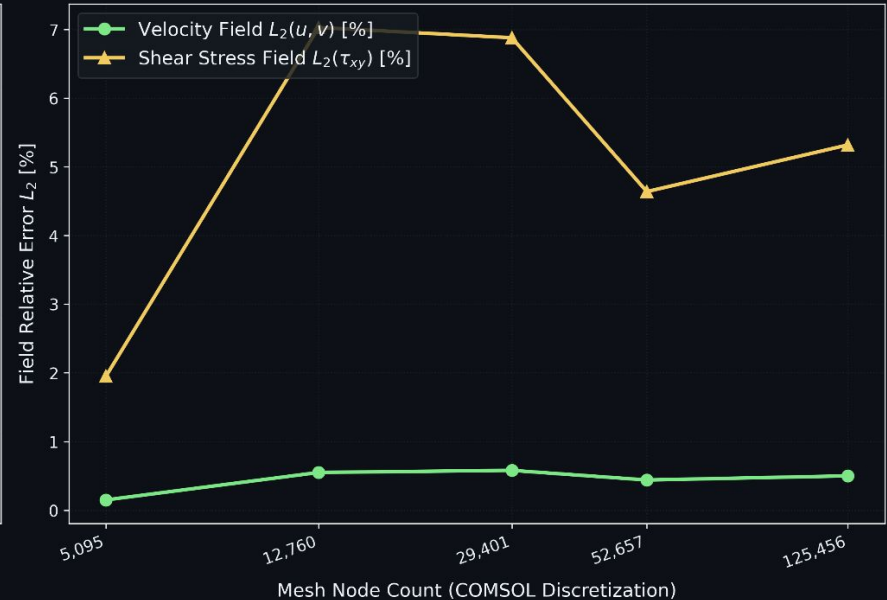
PARAMS:  $\lambda = 0.1$  s,  $\eta p = 0.5$  Pa s

## Grid Convergence Study: Parameter & Field Accuracy vs Mesh Resolution

### Physical Parameters Estimation Error



### Fluid Flow Fields Relative Error ( $L_2$ )



# Validation and Advantages of Transfer Learning

## 01. MOTIVATION

### Why Transfer Learning?

Training a viscoelastic PINN *from scratch* for each new set of physical parameters or operating conditions is:

- **Computationally expensive** in terms of time and resources.
- **Subject to convergence** towards unfavorable local minima.

## 02. STRATEGY

### Implemented Workflow

#### 1. Pre-training

Training on a base case (low  $\lambda$ )

#### 2. Fine-Tuning

Weight transfer towards severe conditions (higher viscoelasticity or kinematic variations).

## 03. RESULTS

### Advantages and Impact

- **Convergence Acceleration**
- **Numerical Stability**
- **High-Efficiency Scalability**

# Phase 2 Inverse Problem Limits

## 01. IDENTIFIABILITY ISSUE

### Identifiability Study on $\eta_s$

Our current setup renders velocity fields insensitive to variations in solvent viscosity ( $\eta_s$ ):

#### Velocity Field Invariance

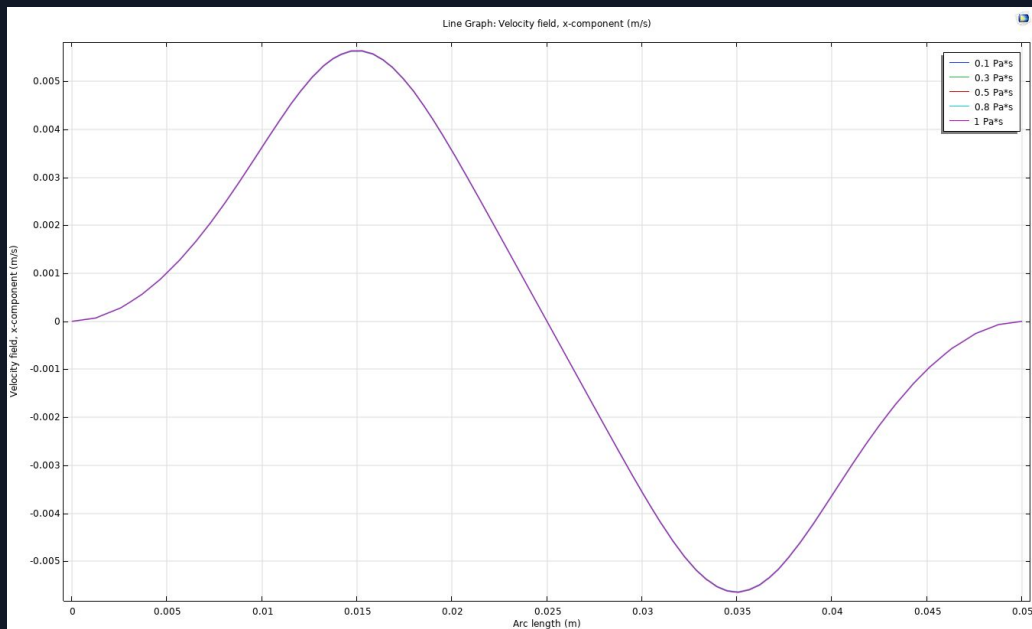
- Changes in  $\eta_s$  produce indistinguishable effects on the velocity profiles.

#### Optimization Bottleneck

- Loss functions struggle to compute gradients for parameter recovery.

## 02. CONVERGENCE ANALYSIS

### Convergence Plot ( $\eta_s$ )



# PTT Inverse Problem Benchmark Results

| Benchmark / Dataset                                                             | $\epsilon_{true} \rightarrow \epsilon_{est}$<br>( $\epsilon$ Error) | $\lambda_{true} \rightarrow \lambda_{est}$ [s]<br>( $\lambda$ Error) | $\mu p_{,true} \rightarrow \mu p_{,est}$ [Pa·s]<br>( $\mu p$ Error) | $L2(u,v)$<br>Kinematics Error |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------|-------------------------------|
| <b>PTT (<math>\lambda=1.0, \epsilon=0.3</math>)</b><br>L1-P0.5-S0.5-A0-E0.3     | 0.300 $\rightarrow$ 1.7 $\times 10^{-4}$<br>-99.94%                 | 1.000 $\rightarrow$ 0.4211<br>-57.89%                                | 0.500 $\rightarrow$ 0.2128<br>-57.44%                               | 0.79%                         |
| <b>PTT (<math>\lambda=0.7, \epsilon=0.5</math>)</b><br>L0.7-P0.95-S0.05-A0-E0.5 | 0.500 $\rightarrow$ 4.4 $\times 10^{-4}$<br>-99.91%                 | 0.700 $\rightarrow$ 0.2700<br>-61.43%                                | 0.950 $\rightarrow$ 0.3687<br>-61.19%                               | 0.88%                         |

## Key Observation:

While individual parameters  $\lambda$ ,  $\mu p$ , and  $\epsilon$  show high estimation errors (systematic underestimation), the **L2 kinematic error remains extremely low (< 0.9%)**, pointing directly to a parameter identifiability issue.

# Proposals to Overcome Setup Limitations

## 01. SOLVENT VISCOSITY ( $\eta_s$ )

### Higher Reynolds Number Regime

Explore a solution space where inertia becomes more relevant (higher  $Re$ ).

#### Objective:

Amplify the physical impact of solvent viscosity on velocity profiles to break sensitivity invariance.

#### Expected Outcome:

Sharpen loss function gradients for accurate recovery of  $\eta_s$ .

## 02. VISCOELASTIC MODEL (PTT)

### Exponential PTT Formulation

Transition to an exponential Phan-Thien-Tanner (PTT) model formulation.

#### Objective:

Overcome mathematical & physical limits tied strictly to the stationary regime setup.

#### Expected Outcome:

Resolve parameter coupling between  $\lambda$ ,  $\tau$ , and  $\eta_s$  for unique parameter identifiability.

# Future Development

## 01. KINEMATICS MODIFICATION

### Flow Field Alteration

Focus on changing the flow field configurations to study non-standard velocity dynamics.

## 02. STOCHASTIC PERTURBATION

### Gaussian Noise Addition

Incorporate Gaussian noise into the synthetic datasets and flow fields to evaluate numerical robustness, regularization stability, and solver resilience under uncertainty.

## 03. CONSTITUTIVE FORMULATION

### Exponential PTT Integration

Transition to exponential PTT (ePTT) instead of linear PTT in ViscoelasticNet.

## 04. NUMERICAL BOUNDS

### Convergence Range Exploration

Conduct thorough parameter sweeps to map out and expand the convergence envelope across varying Weissenberg numbers, Reynolds regimes, and loss weighting setups.

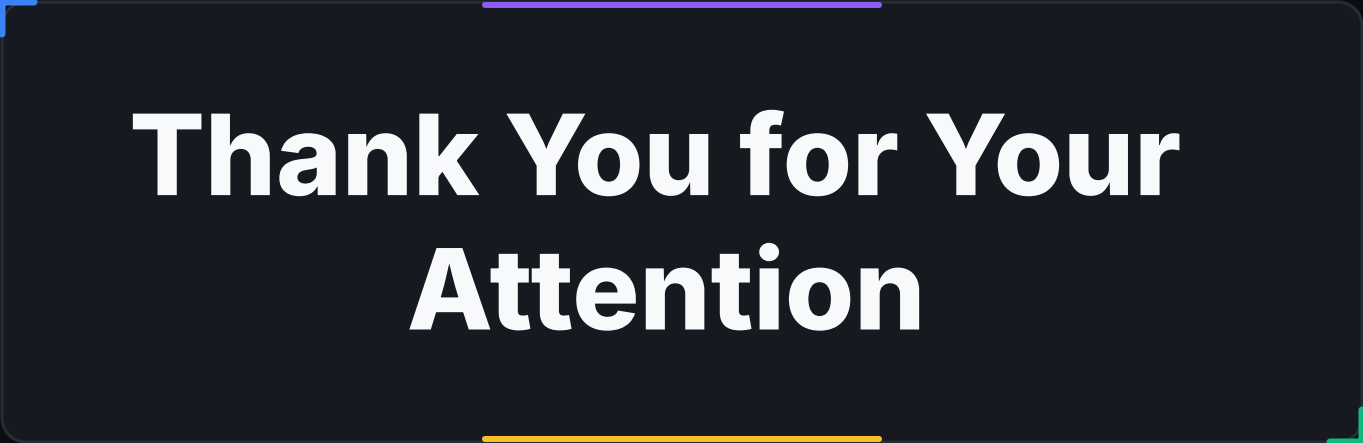
# Latest Literature Development

RECENT PUBLICATION (2026)

## Simulation of Steady Viscoelastic Creeping Flows with Physics-Informed Neural Networks

José X. Matosa, C. Fernandes, M. A. Alves,\*

a CEFT, ALiCE, Faculty of Engineering, University of Porto, Rua Dr. Roberto Frias, s/n 4200-465 Porto, Portugal.



**Thank You for Your  
Attention**

# Governing Equations & Viscoelastic Formulation

Coupling mass conservation, Cauchy momentum balance, and constitutive stress tensor relations

## SYSTEM OF GOVERNING PARTIAL DIFFERENTIAL EQUATIONS

Mass Conservation

$$\nabla \cdot \mathbf{u} = 0$$

Incompressible flow condition

Cauchy Momentum Balance

$$\rho(\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \nabla \cdot \boldsymbol{\tau}$$

Steady-state momentum transport

Constitutive Relation

$$\boldsymbol{\tau} = \boldsymbol{\tau}_s + \boldsymbol{\tau}_p$$

Solvent & polymer stress split

## DIMENSIONLESS PARAMETERS & REGIMES

### Re — Reynolds Number

Characterizes the ratio of inertial forces to viscous forces within the fluid domain, governing flow stability.

### Wi — Weissenberg Number

Quantifies fluid elasticity and polymer relaxation time relative to the characteristic rate of flow deformation.

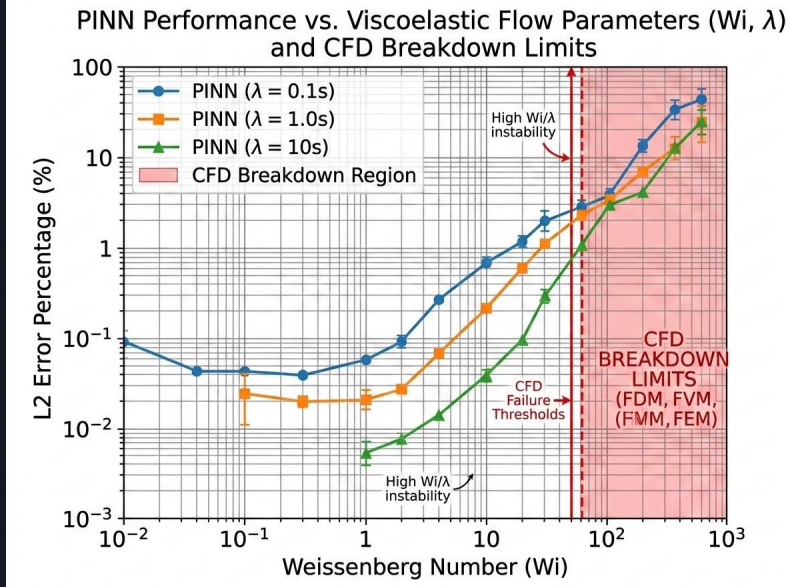
## PHYSICAL DOMAIN & RHEOLOGICAL MODEL

**Flow Specification:** The physical domain models steady-state, incompressible viscoelastic flow under isothermal conditions.

**Stress Response:** Defined by a viscoelastic constitutive model partitioning the total stress tensor into Newtonian solvent and elastic polymer contributions.

**System Coupling:** Momentum balance and non-linear constitutive equations dictate the final flow behavior across the domain.

# High Weissenberg Number Problem Mitigation



Quantitative Benchmark: L2 Error vs. Weissenberg Number ( $Wi$ )

Comparative breakdown highlighting CFD failure thresholds against mesh-free PINN stability.

## Regime Stability & Computational Overhead

### Bypassing Numerical Breakdown

The physics-informed framework demonstrates robust performance in complex regimes, effectively bypassing the High Weissenberg Number Problem.

By eliminating complex meshing requirements for viscoelastic flow domains, the model maintains numerical stability and accuracy even in highly non-linear regimes.

## High-Fidelity Prediction Capability

### Non-Linear Stress & Velocity Kinematics

The mesh-free solver achieves high-fidelity predictions of both velocity kinematics and non-linear stress tensors.

It mitigates the numerical breakdown and steep computational overheads traditionally associated with high elasticity and complex relaxation time behaviors.

# Non so se aggiungere anche questo forse alla fine

## COMPUTATIONAL ENGINEERING IMPLICATIONS

### Take-Home Messages & Method Positioning

#### Core Strengths

- Ideal for solving inverse problems and parameter identification.
- Seamless sparse experimental data assimilation (e.g., PIV velocity fields).
- Quantifies unmeasurable latent physical variables directly.

#### Practical CFD Positioning

- Does **not** aim to replace mature forward CFD solvers (FVM / FEM) for high-fidelity forward predictions.
- Acts as an integrative framework, bridging real experimental data with physical models.

Experimental Data (PIV) + Physical Laws → PINN Integration